

Unit 1

ATOMIC STRUCTURE

71 items · 349 marks

Syllabus sections in this unit

- 1.1 Particles in the atom and atomic radius
- 1.2 Isotopes
- 1.3 Electrons, energy levels and atomic orbitals
- 1.4 Ionisation energy

Unit 1: Atomic structure**1. M/J 26/P21/Q1/b**

(b) Table 1.1 shows information about sulfur and chlorine.

Table 1.1

element	electronic configuration	first ionisation energy /kJ mol ⁻¹
sulfur	[Ne] 3s ² 3p ⁴	1000
chlorine	[Ne] 3s ² 3p ⁵	1260

- (i) Name the orbital that contains the lowest energy electron in an atom of sulfur. Draw the shape of this orbital.

name

shape

[1]

- (ii) Construct an equation to represent the first ionisation energy of chlorine.

[1]

- (iii) Explain the difference in the values of first ionisation energy of sulfur and chlorine in Table 1.1.

[3]

2. M/J 26/P22/Q1/a

1 Oxygen reacts with many elements.

- (a) (i) State the total number of electrons that occupy the 2p sub-shell in an oxygen atom.

[1]

- (ii) Name the orbital that contains the highest energy electron in an oxygen atom. Draw the shape of this orbital.

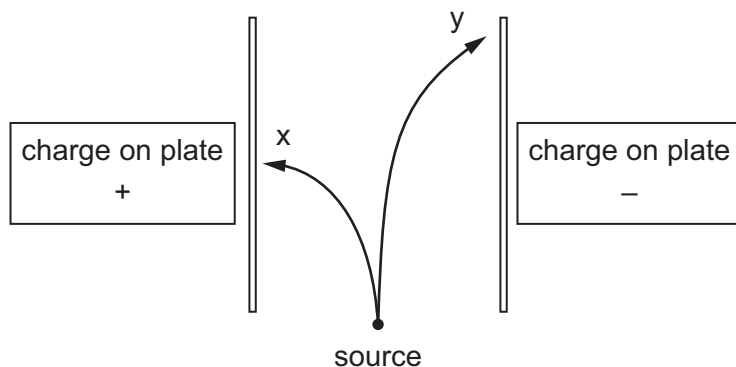
name

shape

[1]

3. M/J 26/P22/Q2/a,b

- 2 (a)** Protons, neutrons and electrons are released with the same velocity from a source. Figure 2.1 shows the movement of two of these types of particle when they travel between oppositely charged plates.

**Figure 2.1**

- (i)** Name each type of particle that produces the deflection shown by curves x and y.

curve x =

curve y =

[1]

- (ii)** Suggest why the particles producing curve y are deflected less than the particles producing curve x.

[1]

- (b)** A student considers the information in Table 2.1 and proposes hypothesis 1.

hypothesis 1 'Atoms with a higher nuclear charge are always larger.'

Table 2.1

atom	atomic number	mass number	atomic radius/nm
A	6	12	0.077
B	17	37	0.099
C	14	28	0.117

- (i)** Identify a pair of atoms from Table 2.1 that supports hypothesis 1.

[1]

- (ii)** Identify a pair of atoms from Table 2.1 that shows hypothesis 1 is false.

[1]

- (iii)** Explain why atom **A** is smaller than atom **C**.

[2]

4. M/J 26/P23/Q2/a

- 2 Group 18 elements exist as colourless gases at room temperature and pressure.

Table 2.1

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

- (a) Use Table 2.1 to describe and explain the variation in atomic radius down Group 18.

[2]

5. M/J 26/P23/Q5/b(i)

- 5 Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (b) The mechanism for this reaction involves three stages.

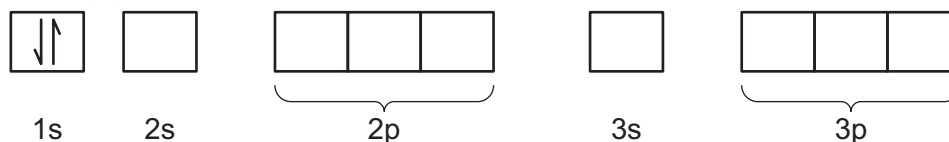
stage 1 initiation

stage 2 propagation

stage 3 termination

During the initiation reaction, a chlorine free radical is produced.

- (i) Complete Figure 5.1 to describe the arrangement of electrons in a chlorine free radical.

**Figure 5.1**

[1]

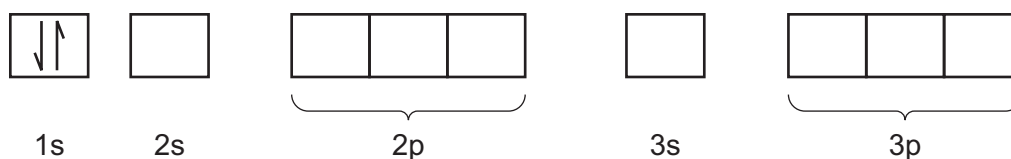
6. M/J 26/P24/Q1/a

- 1 Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

- (a) (i) Complete Table 1.1 to show the number of particles in each atom or ion. [2]
- (ii) Complete Figure 1.1 to show the electronic configuration of an aluminium atom.

**Figure 1.1**

[1]

7. MAR 26/P22/Q1/a

- 1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

- (a) Table 1.1 shows some information about the elements Mg to Ba.

Table 1.1

element	proton number	shorthand electronic configuration	sum of first and second ionisation energies / kJ mol^{-1}	third ionisation energy / kJ mol^{-1}
Mg	12	$[\text{Ne}] 3s^2$	2186	7740
Ca	20	$[\text{Ar}] 4s^2$	1740	4940
Sr	38	$[\text{Kr}] 5s^2$	1608	4120
Ba	56	$[\text{Xe}] 6s^2$	1468	3390

- (i) Give the full electronic configuration of a calcium atom. [1]
- (ii) Explain the trend shown in the sum of the first and second ionisation energies, as proton number increases. [2]
- (iii) Write an equation to represent the third ionisation energy of barium. [1]
- (iv) Explain why the third ionisation energy of magnesium is much greater than the sum of its first and second ionisation energies. [2]

8. M/J 25/P21/Q2/a,b(ii),c,d,e

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

(a) Define isotopes.

[1]

(b) Fig. 2.1 shows the mass spectrum of the sample of iron.

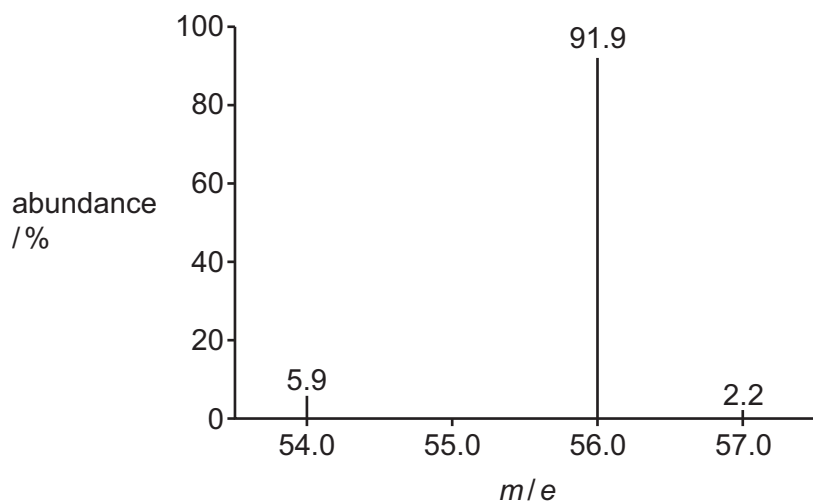


Fig. 2.1

- (ii) Complete Table 2.1 to show the number of protons and nucleons in one atom of ^{56}Fe .

Table 2.1

particle	number of particles in one atom of ^{56}Fe
protons	
nucleons	

[2]

- (c) Deduce the number of pairs of electrons in the shell with principal quantum number $n = 3$ in an Fe atom.

number of pairs of electrons

[1]

- (d) Write an equation to represent the first ionisation energy of iron.

[1]

- (e) Suggest how the value for the first ionisation energy of ^{54}Fe compares to the first ionisation energy of ^{56}Fe . Explain your answer in terms of the factors that affect ionisation energy.

[4]

9. M/J 25/P22/Q2/b,c,d,e

2 A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(b) Deduce the number of pairs of electrons in the 3d sub-shell in an iron(II) ion.

[1]

(c) Sketch the shape of the lowest energy orbital in the shell with principal quantum number $n = 2$.

[1]

(d) Complete Table 2.2 to show information about particles in one ion of $^{56}\text{Fe}^{3+}$.

Table 2.2

particle	number of particles present in one ion of $^{56}\text{Fe}^{3+}$
electrons	
	30

[2]

(e) The atomic radius of iron is $1.26 \times 10^{-10}\text{m}$.

Suggest the change to the radius, if any, after an iron atom reacts to produce an Fe^{3+} ion. Explain your answer.

[2]

10. M/J 25/P23/Q1/a,b

1 The chemical properties of an element are related to the electronic configuration of its atoms.

(a) (i) Give the full electronic configuration of a fluorine atom.

[1]

(ii) Deduce the number of pairs of electrons in the second energy shell, $n = 2$, of an oxygen atom.

[1]

(iii) Draw the shape of the highest energy orbital that contains electrons in an atom of calcium.

[1]

(b) (i) Write an equation to represent the first ionisation energy of sulfur.

[1]

(ii) Explain why the first ionisation energy of sulfur is less than the first ionisation energy of phosphorus.

[2]

(iii) Arrange the three species F^- , Ne and Na^+ in order of increasing radius.

Explain your answer.

smallest radius < < largest radius

[4]

11. M/J 25/P24/Q1/c

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(c) The **third** ionisation energies of the Period 3 elements are shown in Fig. 1.2.

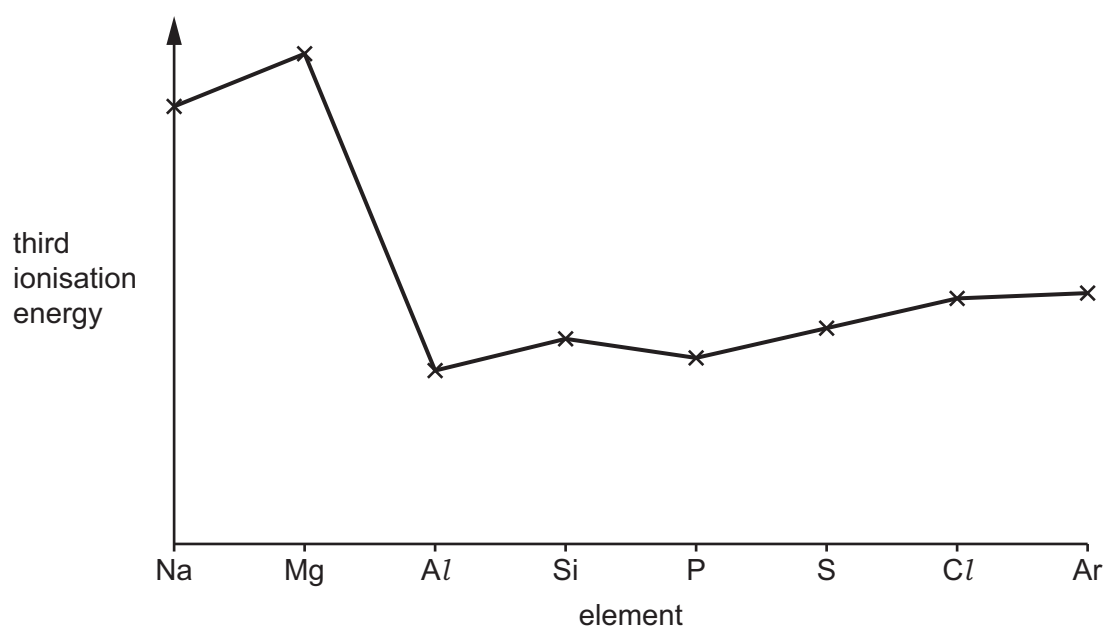


Fig. 1.2

- (i) Write an equation, including state symbols, to represent the **third** ionisation energy of argon.

[2]

- (ii) The differences in the values for third ionisation energy shown in Fig. 1.2 are due to differences in the strength of attraction between the nucleus and the outer electron of each ion.

State **two** factors that affect the strength of attraction between the nucleus and the outer electron.

1

2

[2]

- (iii) Use Fig. 1.2 to suggest the most significant factor that determines the size of attraction between the nucleus and the outer electron. Explain your answer.

[2]

- (iv) In the third ionisation energy of argon, the ion produced has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$.

Complete Fig. 1.3 to show the arrangement of electrons in the orbitals of this ion.

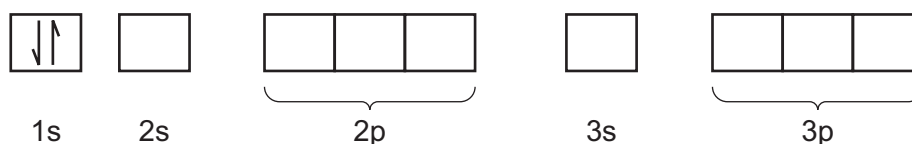


Fig. 1.3

[1]

12. MAR 25/P22/Q2/b,c

- (b) The Group 2 elements form stable $2+$ cations.

- (i) State and explain the variation in ionic radius of the Group 2 elements down the group.

[2]

- (ii) Table 2.1 shows successive ionisation energy values for beryllium, Be.

Table 2.1

	1st	2nd	3rd	4th
ionisation energy / kJ mol^{-1}	900	1760	14 800	21 000

Use Table 2.1 to state and explain:

- the general trend in these values
- the significance of the large difference between the 2nd and 3rd ionisation energies.

[2]

- (c) All the Group 2 elements except beryllium have more than one stable isotope.

- (i) Beryllium exists as the single isotope ${}^9_4\text{Be}$.

Describe the distribution of mass within an atom of ${}^9_4\text{Be}$.

[2]

- (ii) Complete Table 2.2 to show the numbers of protons and neutrons in the isotopes of magnesium.

Table 2.2

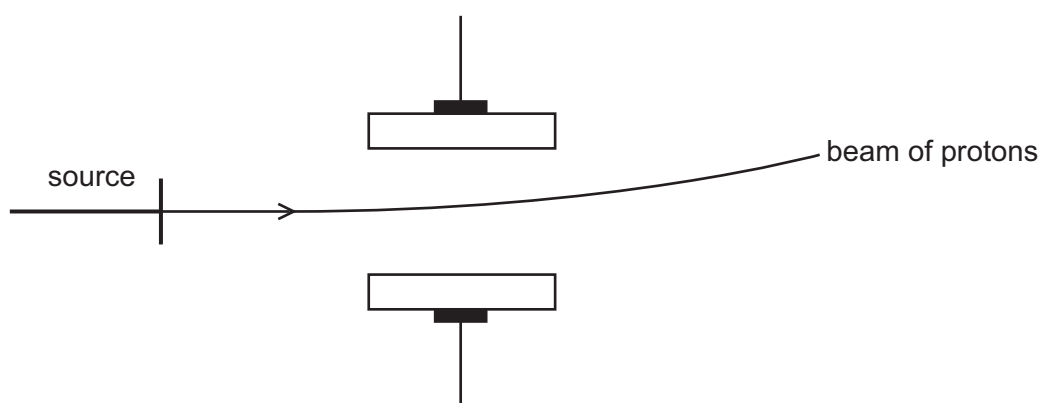
isotope	number of protons	number of neutrons
magnesium-24		
magnesium-25		
magnesium-26		

[2]

- (iii) Fig. 2.1 shows the behaviour of a beam of protons in an electric field.

Complete Fig. 2.1 to show the behaviour of separate beams of neutrons and electrons in the same electric field.

Label your diagram clearly. Assume that the beams of each particle are moving at the same velocity.

**Fig. 2.1**

[3]

13. O/N 24/P21/Q1/a,b

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is $[\text{Ar}]3d^74s^2$.

(i) Identify what is meant by $[\text{Ar}]$ by giving its full electronic configuration.

[1]

(ii) The lowest-energy electrons in cobalt are in the 1s orbital.

Draw the shape of a 1s orbital.

[1]

(iii) Deduce the number of unpaired electrons in a cobalt atom.

[1]

(b) Table 1.1 gives some details of the stable naturally occurring isotopes of rhodium and iridium.

Table 1.1

isotope	number of protons	number of neutrons	total number of electron shells
$^{103}_{45}\text{Rh}$		58	
$^{191}_{77}\text{Ir}$			6
$^{193}_{77}\text{Ir}$			6

Complete Table 1.1.

[3]

14. O/N 24/P21/Q3/c

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of silicon, Si.

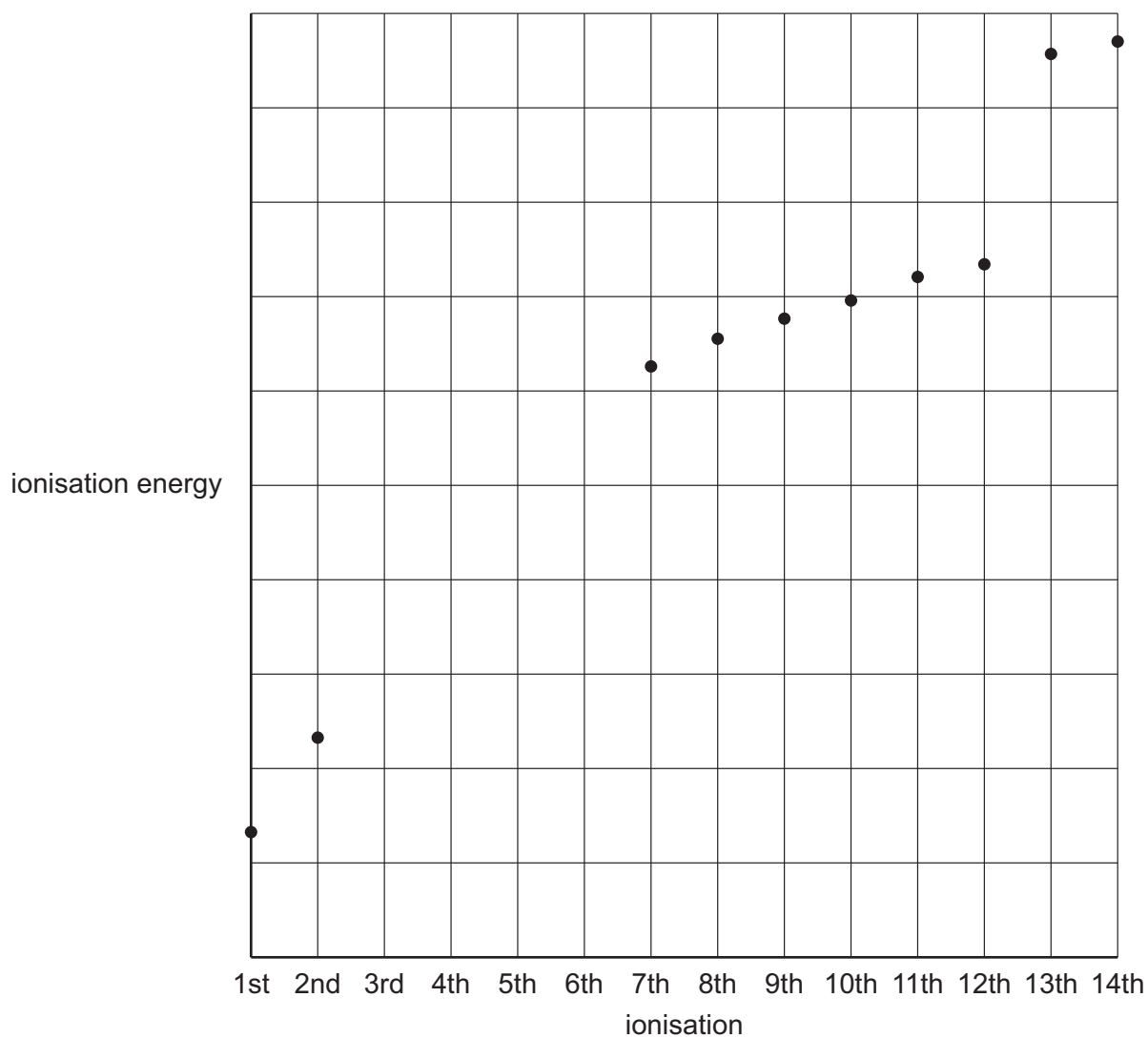


Fig. 3.1

(i) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of Si.

[2]

(ii) Construct an equation to represent the second ionisation energy of Si.

[1]

15. O/N 24/P22/Q1/a,b(i)

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of vanadium in the ground state is $[\text{Ar}]3d^34s^2$.

(i) State what is meant by the term ground state.

[1]

(ii) Show the electronic configuration of vanadium using electrons in boxes notation.

$[\text{Ar}]$

[1]

(iii) Deduce the total number of electrons in the p sub-shells of a vanadium atom.

[1]

(b) Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

(i) Complete Table 1.1.

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
$^{93}_{41}\text{Nb}$	92.91		
$^{181}_{73}\text{Ta}$	180.95		

[2]

16. O/N 24/P22/Q3/c

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of phosphorus, P.

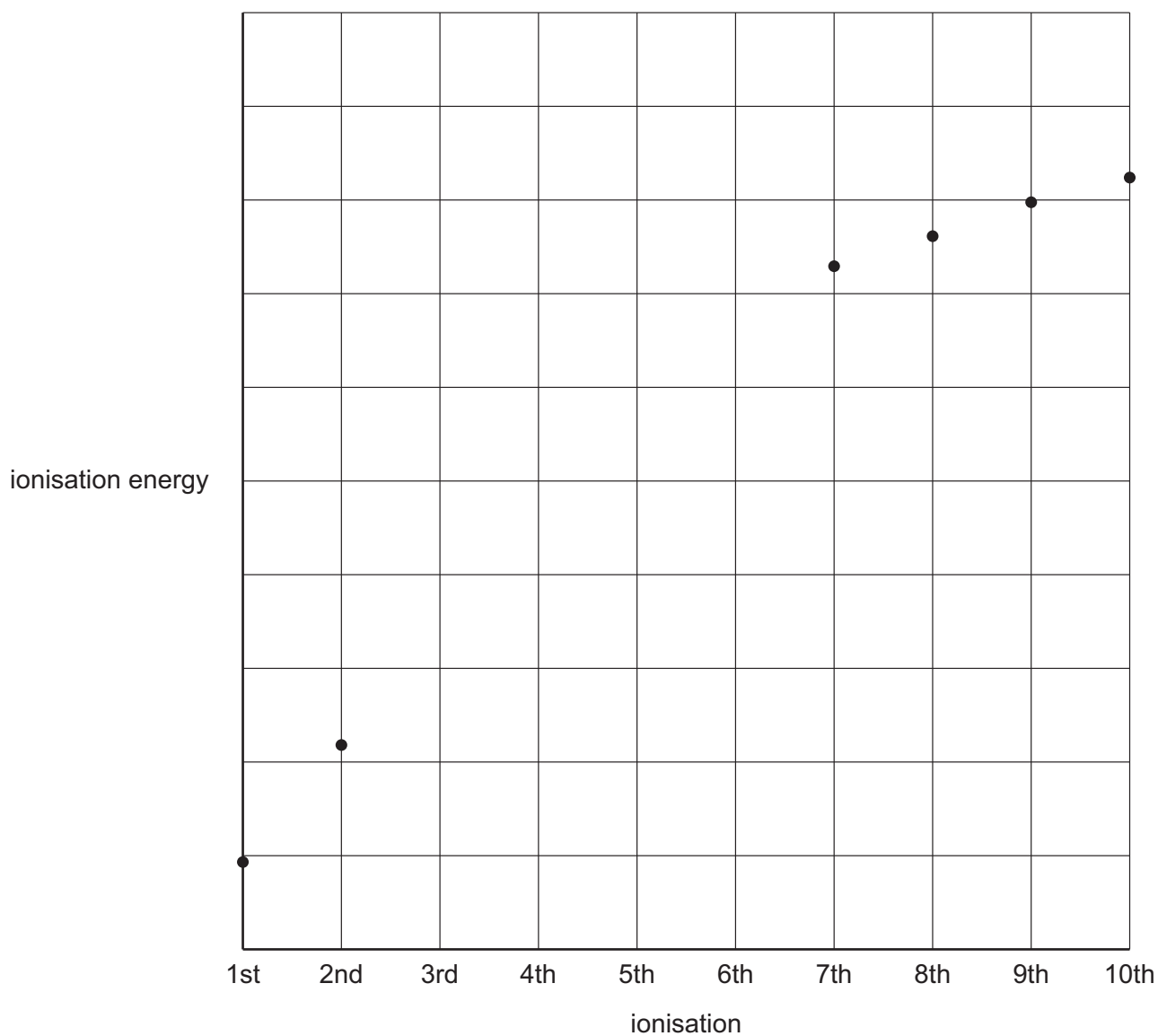


Fig. 3.1

(i) Construct an equation to represent the third ionisation energy of P.

[1]

(ii) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of P.

[2]

17. O/N 24/P23/Q1/a,b

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is $[\text{Ar}]3d^74s^2$.

(i) Identify what is meant by $[\text{Ar}]$ by giving its full electronic configuration.

[1]

(ii) The lowest-energy electrons in cobalt are in the 1s orbital.

Draw the shape of a 1s orbital.

[1]

(iii) Deduce the number of unpaired electrons in a cobalt atom.

[1]

(b) Table 1.1 gives some details of the stable naturally occurring isotopes of rhodium and iridium.

Table 1.1

isotope	number of protons	number of neutrons	total number of electron shells
$^{103}_{45}\text{Rh}$		58	
$^{191}_{77}\text{Ir}$			6
$^{193}_{77}\text{Ir}$			6

Complete Table 1.1.

[3]

18. O/N 24/P23/Q3/c

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of silicon, Si.

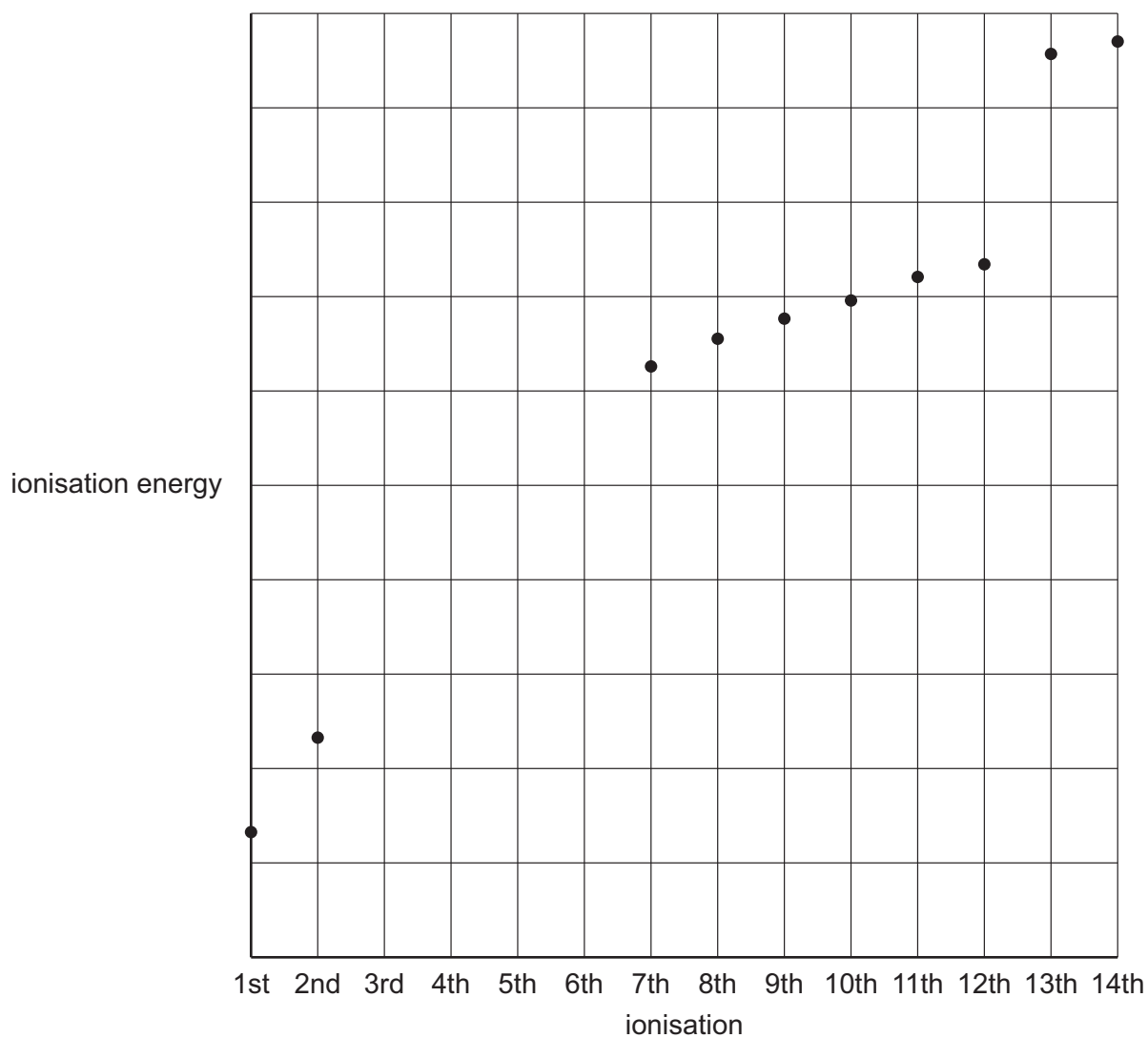


Fig. 3.1

(i) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of Si.

[2]

(ii) Construct an equation to represent the second ionisation energy of Si.

[1]

19. M/J 24/P22/Q1/a,b

- 1 (a)** Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

- (b)** State and explain the difference in the ionic radius of Al^{3+} compared to Mg^{2+} .

[3]

20. M/J 24/P22/Q3/c

- (c)** Determine the full electronic configuration of Fe^{3+} .

[1]

21. O/N 23/P21/Q1/a,b

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and Cl. [2]

- (ii) Construct an equation to represent the first ionisation energy of P.

[1]

- (iii) Three possible values for the first ionisation energy of S are given.

1000 kJ mol⁻¹

1160 kJ mol⁻¹

1320 kJ mol⁻¹

Circle the correct value.

Explain your choice by comparing your chosen value to those of P and Cl.

[4]

- (b) P³⁻, S²⁻ and Cl⁻ have the same number of electrons.

- (i) Give the full electronic configuration of P³⁻.

[1]

- (ii) State the trend in ionic radius shown by P³⁻, S²⁻ and Cl⁻.

Explain your answer.

[2]

22. O/N 23/P22/Q1/b

also Unit 9 (1 mark), Unit 3 (1 mark); this unit 7 marks

- 1** The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

- (b)** Table 1.1 shows some properties of the elements Si to S.

The first ionisation energy of P is **not** shown.

Table 1.1

property	Si	P	S
total number of electrons in s subshells			
total number of electrons in p subshells			
first ionisation energy/kJ mol ⁻¹	786		1000
formula of most common chloride	SiCl ₄	PCl ₅	SCl ₂

- (i)** Complete Table 1.1 to show the total number of s and p electrons in an atom of Si, P and S.

[2]

- (ii)** Construct an equation to represent the first ionisation energy of Si.

[1]

- (iii)** Three possible values for the first ionisation energy of P are given.

619 kJ mol⁻¹

893 kJ mol⁻¹

1060 kJ mol⁻¹

Circle the correct value.

Explain your choice, including a comparison of your chosen value to those of Si and S.

[4]

- (iv)** SiCl₄ and PCl₅ each react with water, forming misty fumes.

Identify the chemical responsible for the misty fumes.

[1]

- (v)** Predict the shape of the SCl₂ molecule.

[1]

23. O/N 23/P23/Q1/a,b

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and Cl. [2]

- (ii) Construct an equation to represent the first ionisation energy of P.

[1]

- (iii) Three possible values for the first ionisation energy of S are given.

1000 kJ mol⁻¹

1160 kJ mol⁻¹

1320 kJ mol⁻¹

Circle the correct value.

Explain your choice by comparing your chosen value to those of P and Cl.

[4]

- (b) P³⁻, S²⁻ and Cl⁻ have the same number of electrons.

- (i) Give the full electronic configuration of P³⁻.

[1]

- (ii) State the trend in ionic radius shown by P³⁻, S²⁻ and Cl⁻.

Explain your answer.

[2]

24. M/J 23/P21/Q1/a,b,c,d

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(a) Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
^{130}Te			

[3]

(b) Identify the sub-shell in an atom of Te that contains electrons with the lowest energy.

[1]

(c) Construct an equation to represent the first ionisation energy of Te.

[1]

(d) (i) The radius of Te ions decreases after each successive ionisation.

State **two** factors that are responsible for the increase in the first six ionisation energies of Te.

[2]

(ii) Sketch a graph in Fig. 1.1 to show the trend in the first **seven** ionisation energies of Te.

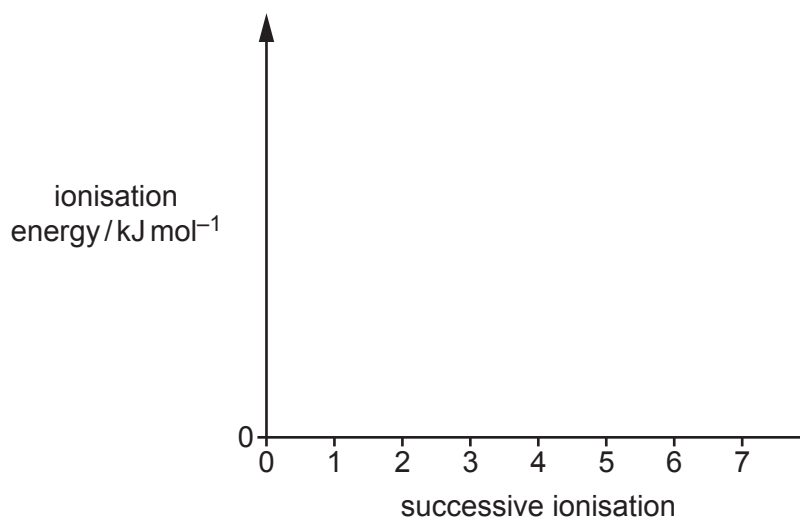


Fig. 1.1

[2]

25. M/J 23/P23/Q1/d(iii)

- 1 Copper is used in electrical equipment. It has a melting point of 1085 °C.
- (d) When KI(aq) is added to CuSO₄(aq) the blue-coloured solution turns brown and a white precipitate of CuI(s) is seen.
- The reaction between copper ions and iodide forms only two products.
- (iii) State the full electronic configuration of Cu²⁺.

[1]

26. MAR 23/P22/Q1/c

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

- (c) (i) Write an equation that represents the first ionisation energy of H.
- (ii) Explain why there is no information given in Table 1.1 for the second ionisation energy of H.
- (iii) Give the full electronic configuration of S²⁺(g).

[1]

[1]

[1]

27. O/N 22/P21/Q1/a(i),b

- 1 Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

- (i) Complete Table 1.1 to show the numbers of protons, neutrons and electrons in the two stable isotopes of gallium.

Table 1.1

isotope	number of protons	number of neutrons	number of electrons
^{69}Ga			
^{71}Ga			

[2]

(b) Potassium also has two stable isotopes. Both isotopes have the same chemical properties.

- (i) Explain why both isotopes of potassium have the same chemical properties.

[1]

- (ii) State the full electronic configuration of an atom of potassium.

[1]

- (iii) The first, second and third ionisation energies of potassium are 418, 3070 and 4600 kJ mol⁻¹, respectively.

Use this information to explain why potassium is in Group 1.

[2]

28. O/N 22/P22/Q1/b

- 1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(b) Complete Table 1.1 to show the total numbers of protons and electrons in the molecular ions NH_4^+ , CO_3^{2-} and PO_4^{3-} .

Table 1.1

molecular ion	total number of protons	total number of electrons
NH_4^+		
CO_3^{2-}		
PO_4^{3-}		

[3]

29. O/N 22/P23/Q1/a(i),b

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

- (i)** Complete Table 1.1 to show the numbers of protons, neutrons and electrons in the two stable isotopes of gallium.

Table 1.1

isotope	number of protons	number of neutrons	number of electrons
^{69}Ga			
^{71}Ga			

[2]

(b) Potassium also has two stable isotopes. Both isotopes have the same chemical properties.

- (i)** Explain why both isotopes of potassium have the same chemical properties.

[1]

- (ii)** State the full electronic configuration of an atom of potassium.

[1]

- (iii)** The first, second and third ionisation energies of potassium are 418, 3070 and 4600 kJ mol⁻¹, respectively.

Use this information to explain why potassium is in Group 1.

[2]

30. M/J 22/P21/Q1/a,d,e(iii)

- 1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

- (a) Identify the highest energy orbital which contains electrons in a calcium atom. Sketch the shape of this orbital.

identity of highest energy orbital in Ca

shape

[1]

- (d) (i) $^{25}_{12}\text{Mg}$ is an isotope of magnesium.

Determine the number of protons and neutrons in an atom of $^{25}_{12}\text{Mg}$.

number of protons

number of neutrons

[1]

- (ii) State the full electronic configuration of an atom of $^{25}_{12}\text{Mg}$.

[1]

- (e) A sample of magnesium contains three isotopes, ^{25}Mg , ^{26}Mg and **X**.

The percentage abundance of the three isotopes is shown in Table 1.1.

Table 1.1

isotope of Mg	mass/a.m.u.	percentage abundance/%
X		78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

- (iii) State **one** similarity and **one** difference in the properties of these isotopes of magnesium. Explain your answer.

[2]

31. M/J 22/P22/Q4/b(iii)

(b) Halothane is an anaesthetic.

halothane

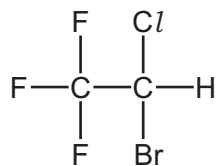


Fig. 4.1

(iii) Complete Fig. 4.2 to show the arrangement of electrons in a bromine atom using the electrons in boxes notation.

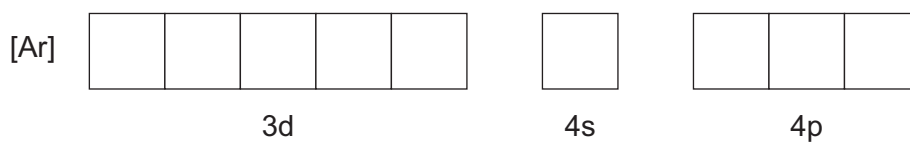


Fig. 4.2

[1]

32. M/J 22/P23/Q1/a,b,c,d,e

- 1 (a) Define first ionisation energy.

[2]

- (b) Successive ionisation energies for element **A** are shown in Table 1.1.

Table 1.1

ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th
ionisation energy/kJ mol ⁻¹	1310	3390	5320	7450	11 000	13 300	71 000	84 100

Use Table 1.1 to deduce the group of the Periodic Table that **A** belongs to. Explain your answer.

Group

[1]

- (c) Across Period 3 there is a general trend for first ionisation energies to increase due to the increase in attraction between the nucleus and the outer electron.

Explain why the first ionisation energy of sulfur is less than the first ionisation energy of phosphorus.

[2]

- (d) In an Al^{2+} ion the nuclear attraction for the outer electron is stronger than in an atom of Na.

Compare the electronic structures of Al^{2+} and an atom of Na and explain why the third ionisation energy of aluminium is greater than the first ionisation energy of sodium.

[2]

- (e) An isotope of copper has a relative isotopic mass of 65.

Complete Table 1.2 for an atom of copper-65.

Table 1.2

	atomic number	nucleon number	number of neutrons	electronic arrangement
copper-65				

[3]

33. MAR 22/P22/Q1/a,b,c

1 Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

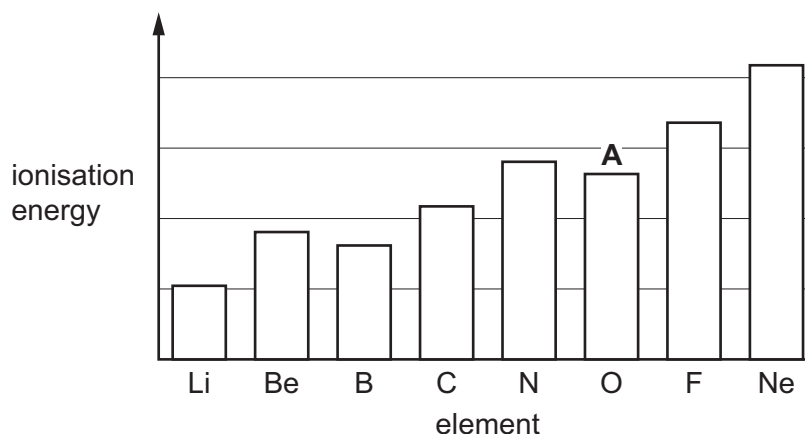


Fig. 1.1

- (a) Construct an equation to represent the **first** ionisation energy of oxygen. Include state symbols.

[1]

- (b) (i) State and explain the general trend in first ionisation energies across Period 2.

[3]

- (ii) Explain why ionisation energy **A** in Fig. 1.1 does **not** follow the general trend in first ionisation energies across Period 2.

[2]

- (c) Element **E** is in Period 3 of the Periodic Table. The first eight ionisation energy values of **E** are shown in Table 1.1.

Table 1.1

ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th
ionisation energy/kJ mol ⁻¹	577	1820	2740	11 600	14 800	18 400	23 400	27 500

Deduce the full electronic configuration of **E**.
Explain your answer.

full electronic configuration of **E** =

explanation

[3]

34. O/N 21/P22/Q2/a(ii,iii,v)

2 (a) Table 1 gives physical data for some of the Period 3 elements.

Table 1

atomic number, Z	11	12	13	14	15	16	17
bonding present in element	M						C
first ionisation energy / kJ mol ⁻¹	494	736	577	786	1060	1000	1260
maximum oxidation number							+7
anionic radius / nm	–	–	–	0.271	0.212	0.184	0.181

(ii) Explain the difference between the first ionisation energies of the elements with atomic numbers 11 and 17.

[2]

(iii) Explain the difference between the first ionisation energies of the elements with atomic numbers 15 and 16.

[2]

(v) Explain the variation in anionic radius for the elements with atomic numbers 14 to 17.

[2]

35. M/J 21/P22/Q1/f(ii)

1 A Group 2 metal combines with bromine to form a crystalline solid, **MBr₂**.

Excess aqueous AgNO₃ is added to a solution of **MBr₂** and a precipitate forms. The mixture is filtered. The precipitate is dried and the mass of the precipitate is recorded.

(f) Separate 1.0 g samples of three different magnesium salts are tested in order to identify the anion present in each sample.

(ii) Complete the electron configuration of the magnesium cation present in these salts.

1s²

[1]

36. MAR 21/P22/Q2/b,g(iii)

2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

(b) Chlorine has the highest first ionisation energy of the Period 3 elements Na to Cl .

(i) Construct an equation for the first ionisation energy of chlorine.

Include state symbols.

[1]

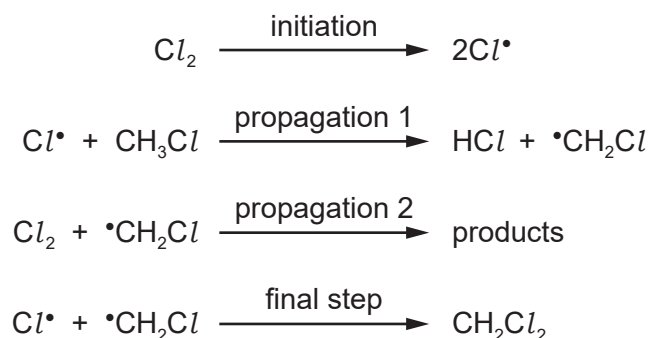
(ii) Explain the general increase in the first ionisation energies of the Period 3 elements.

[2]

(g) Dichloromethane, CH_2Cl_2 , is widely used as an organic solvent.

CH_2Cl_2 can be prepared by reacting CH_3Cl and Cl_2 at room temperature.

The reaction proceeds via several steps, as shown.



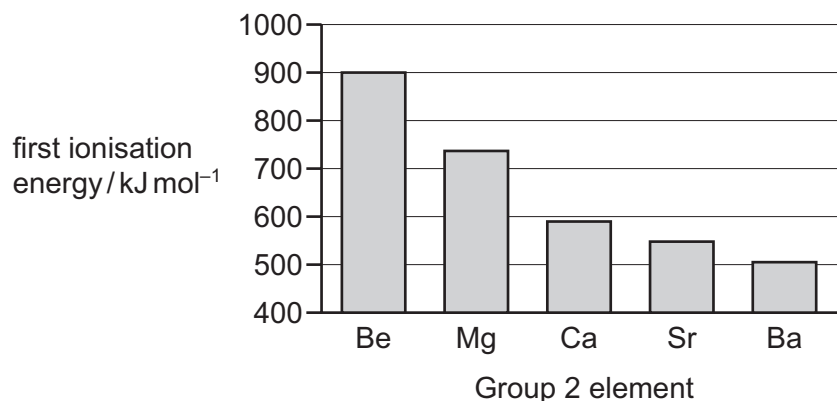
(iii) Give the electronic configuration of $\text{Cl}^\bullet$.

$1s^2$

[1]

37. O/N 20/P21/Q1/a,b,c

- 1 The graph shows the first ionisation energies of some of the elements in Group 2.



- (a) Write an equation for the first ionisation energy of Mg.

Include state symbols.

[1]

- (b) Explain the observed trend in first ionisation energies down Group 2.

[3]

- (c) The second ionisation energy of Be is 1757 kJ mol⁻¹.

Explain why the second ionisation energy of Be is higher than the first ionisation energy of Be.

[2]

38. O/N 20/P21/Q3/a(iii)

- 3 The reducing agent LiAlH₄ can be synthesised by reacting aluminium chloride with lithium hydride, LiH.

- (iii) Lithium hydride contains the ions Li⁺ and H⁻.

State the electronic configuration of these two ions.

Li⁺

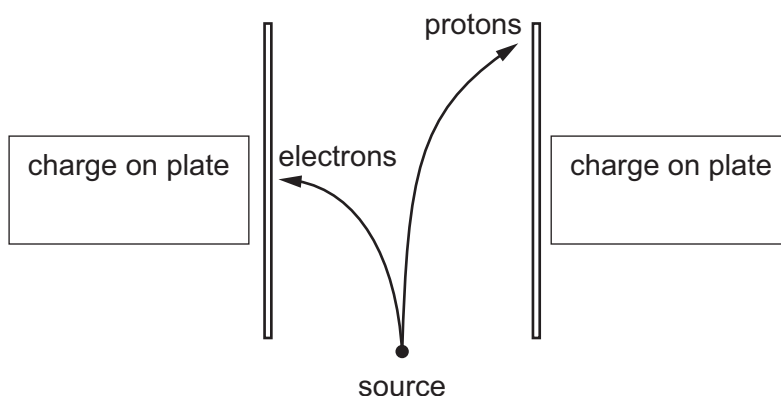
H⁻

[1]

39. O/N 20/P22/Q1/a,b(i,ii,iii),c

- 1 Atoms contain the subatomic particles electrons, protons and neutrons. Protons and electrons were discovered by observations of their behaviours in electric fields.

- (a) The diagram shows the behaviour of separate beams of electrons and protons in an electric field.



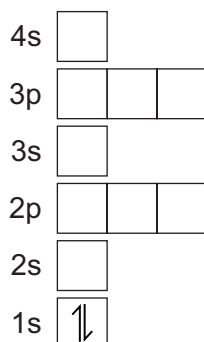
- (i) Complete the diagram with the relative charge of each of the electrically charged plates. [1]
- (ii) On the diagram, draw a line to show how a separate beam of neutrons from the same source behaves in the same electric field. [1]
- (b) Electrons in atoms up to $_{36}\text{Kr}$ are distributed in s, p and d orbitals.

- (i) State the number of occupied orbitals in an isolated atom of $_{36}\text{Kr}$.

type of orbital	s	p	d
number of orbitals			

[3]

- (ii) Complete the diagram to show the number and relative energies of the electrons in an isolated atom of $_{14}\text{Si}$.



[2]

- (iii) The diagram shows a type of orbital.



State the total number of electrons that exist in all orbitals of this type in an atom of $_9\text{F}$.

[1]

- (c) *Hydron* is a general term used to represent the ions $^1_1\text{H}^+$, $^2_1\text{H}^+$ and $^3_1\text{H}^+$.

State, in terms of subatomic particles in the nucleus, what is the same about each of these ions and what is different.

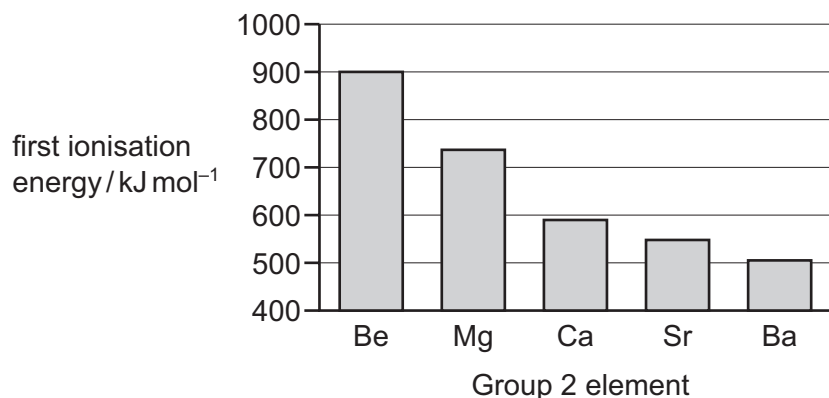
same

different

[1]

40. O/N 20/P23/Q1/a,b,c

- 1 The graph shows the first ionisation energies of some of the elements in Group 2.



- (a) Write an equation for the first ionisation energy of Mg.

Include state symbols.

[1]

- (b) Explain the observed trend in first ionisation energies down Group 2.

[3]

- (c) The second ionisation energy of Be is 1757 kJ mol⁻¹.

Explain why the second ionisation energy of Be is higher than the first ionisation energy of Be.

[2]

41. O/N 20/P23/Q3/a(iii)

- 3 The reducing agent LiAlH₄ can be synthesised by reacting aluminium chloride with lithium hydride, LiH.

- (iii) Lithium hydride contains the ions Li⁺ and H⁻.

State the electronic configuration of these two ions.

Li⁺

H⁻

[1]

42. M/J 20/P21/Q1/a(i)

- 1 Gallium is a metal in Group 13 of the Periodic Table.

- (a) There are two stable isotopes of gallium, ⁶⁹Ga and ⁷¹Ga.

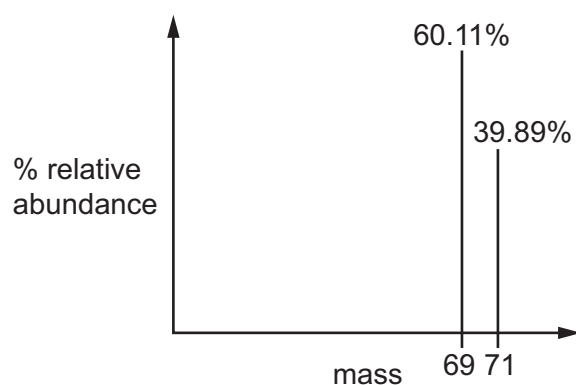
- (i) State, with reference to subatomic particles, how the isotopes ⁶⁹Ga and ⁷¹Ga differ from each other.

[1]

43. M/J 20/P22/Q1/c

- 1** Gallium is an element in Group 13.

A sample of gallium is analysed using a mass spectrometer. The mass spectrum produced is shown.



- (c)** Complete the table which describes a gaseous atom of gallium.

isotope	nucleon number	total number of electrons in lowest energy level	type of orbital which contains the electron in the highest energy level
^{71}Ga			

[3]

44. O/N 19/P22/Q1/a

- 1 In the Periodic Table, the p block contains elements whose outer electrons are found in the p subshell.

(a) Elements in the p block show a general increase in first ionisation energy as the atomic number increases.

(i) Draw the shape of a p orbital.

[1]

(ii) Write an equation to show the first ionisation energy of silicon.

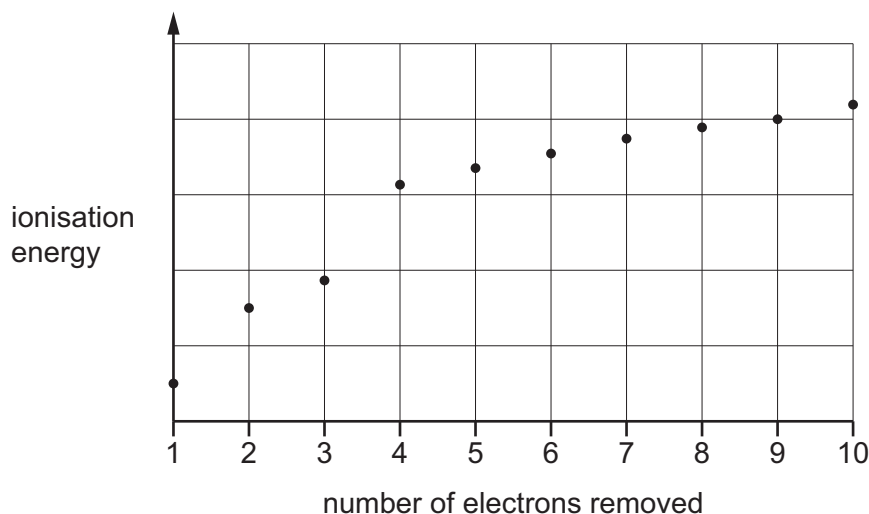
[1]

(iii) Explain why there is a general increase in first ionisation energies of the elements across Period 3.

[2]

(iv) Element **A** is in the p block.

The graph shows the successive ionisation energies for the removal of the first ten electrons of **A**.



State and explain the group of the Periodic Table that element **A** belongs to.

group number

explanation

[2]

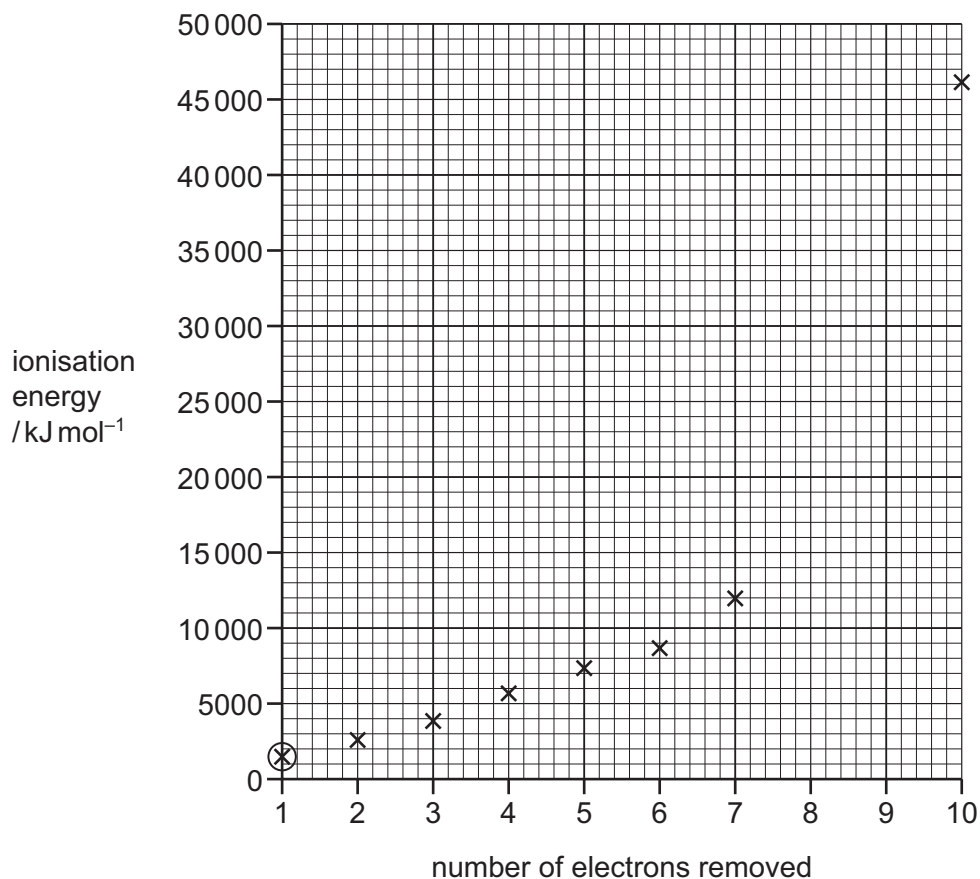
45. M/J 19/P21/Q3/a,b,c

- 3 (a)** Construct an equation for the **second** ionisation energy of argon.

[1]

- (b)** The graph shows successive ionisation energies for the element argon.

Complete the graph with predictions for the eighth and ninth ionisation energies of argon. Use a cross (×) for each data point. [2]



- (c)** The energy value required to remove the first electron from an atom of argon is circled on the graph.

Sketch the shape of the orbital that contains this electron.

[1]

46. M/J 19/P23/Q1/a,c,d

1 (a) A sample contains three different types of atom: ${}^{40}_{18}\text{Ar}$, ${}^{40}_{19}\text{K}$ and ${}^{40}_{20}\text{Ca}$.

(i) State fully, in terms of the numbers of subatomic particles, what these three atoms have in common.

[1]

(ii) State fully, in terms of the numbers of **all** subatomic particles, how these three atoms **differ** from each other.

[1]

(c) The electronic configuration of a sulfur atom is $1s^2 2s^2 2p^6 3s^2 3p^4$.

(i) Identify which orbital in a sulfur atom has the lowest energy.

[1]

(ii) Sketch the shape of a p orbital.

[1]

(iii) During the process of ionisation a sulfur atom loses an electron.



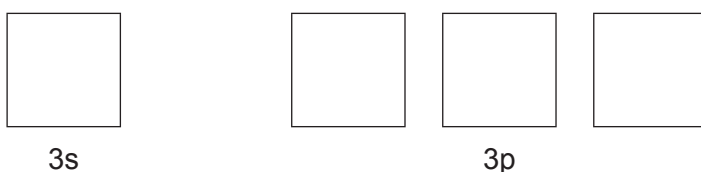
Identify the orbital from which this electron is removed. Explain your answer.

orbital

explanation

[2]

(d) (i) Complete the diagram to show the arrangement of electrons within the third shell of a phosphorus atom.



[1]

(ii) Explain why the first ionisation energy of sulfur is less than that of phosphorus.

[2]

47. MAR 19/P22/Q2/b(i)

- 2 The elements in Group 17 of the Periodic Table are called the halogens. They form stable compounds with both metals and non-metals.

The table gives some data about F_2 , HCl and CaF_2 .

	F_2	HCl	CaF_2
boiling point/K	85	188	2773
relative formula mass	38.0	36.5	78.1

- (b) (i) Complete the electronic configuration of a chloride ion.

$1s^2$

[1]

48. O/N 18/P21/Q1/a(i)

- 1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

- (a) (i) Give the full electronic configuration of Fe^{2+} .

$1s^2$

[1]

49. O/N 18/P22/Q1/a,c

- 1 The model of the nuclear atom was first proposed by Ernest Rutherford. He developed this model on the basis of results obtained from an experiment using gold metal foil.

- (a) Complete the table with information for two of the particles in an atom of ^{197}Au .

particle	relative mass	relative charge	location within atom	total number in an atom of ^{197}Au
electron	0.0005	-1		79
neutron			nucleus	

[4]

- (c) A sample of gold found in the earth consists of only one isotope.

- (i) Explain what is meant by the term *isotopes*.

[2]

- (ii) A different sample of gold contains more than one isotope.

Suggest why this different sample of gold has the same **chemical** properties as the sample found in the earth.

[1]

50. O/N 18/P22/Q2/f

Data Booklet needed

2 The table gives some data for elements in the third period and some of their compounds.

element	Na	Mg	Al	Si	P	S
type of bonding	metallic				covalent	covalent
formula of oxide					P ₄ O ₁₀	SO ₂
formula of chloride	NaCl	MgCl ₂				SCl ₂

- (f) The elements in the third period show a general increase in their first ionisation energies from left to right.

Identify **two** pairs of successive elements in the third period that do **not** agree with this statement.

For each pair, explain why the change in ionisation energy does **not** agree with this statement.

Use of the Data Booklet may help you to answer this question.

pair 1

explanation

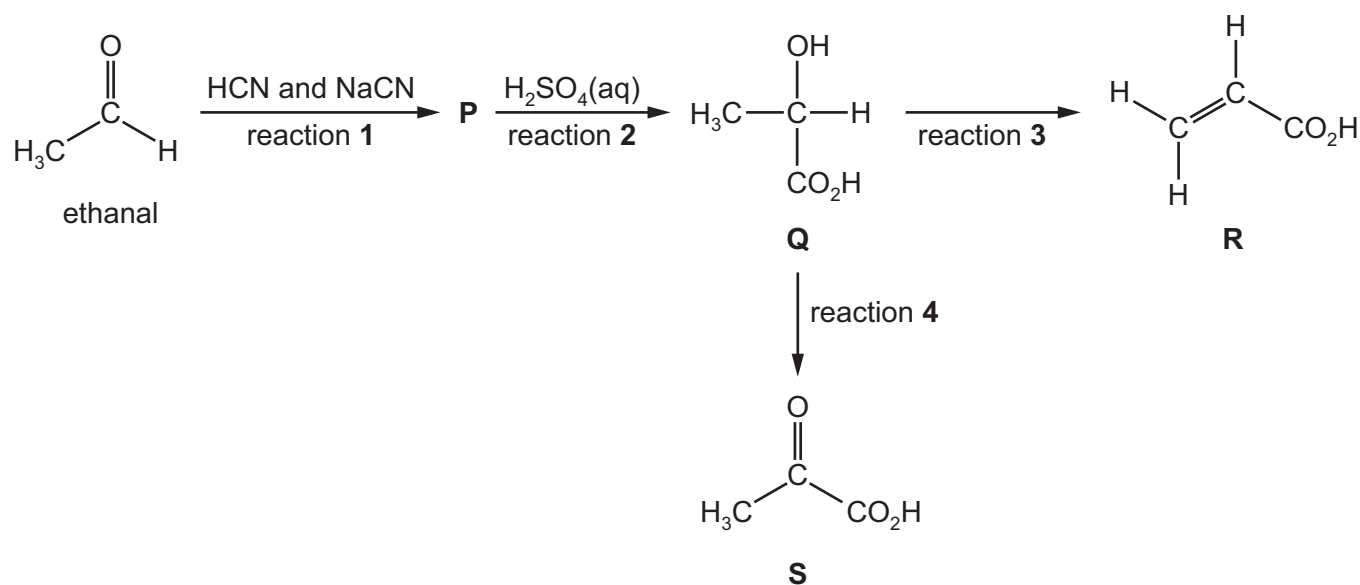
pair 2

explanation

[4]

51. O/N 18/P22/Q4/e(ii,iii)

4 The diagram shows a reaction sequence starting from ethanal.



(ii) Give the electronic configuration of the Na^+ cation.

$1s^2$

[1]

(iii) Suggest why ionic radius increases down Group 1.

[1]

52. O/N 18/P23/Q1/a(i)

1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

(a) (i) Give the full electronic configuration of Fe^{2+} .

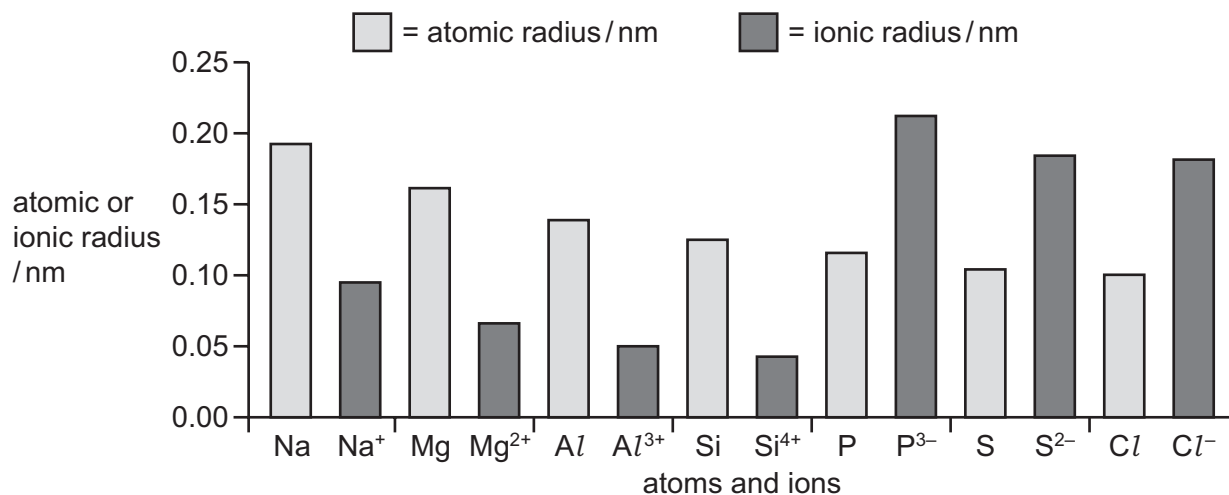
$1s^2$

[1]

53. M/J 18/P21/Q3/a(ii,iii),b

3 The elements in the third period exhibit periodicity in both their chemical and physical properties.

(a) A graph of the atomic and ionic radii across the third period is shown.



(ii) Explain why, for sodium to silicon, the ionic radii are less than the atomic radii.

[1]

(iii) Explain why, for phosphorus to chlorine, the ionic radii are greater than the atomic radii.

[2]

(b) The first ionisation energies of the elements across the third period show a general increase.

Aluminium and sulfur do **not** follow this general trend.

(i) Explain why aluminium has a lower first ionisation energy than magnesium.

[2]

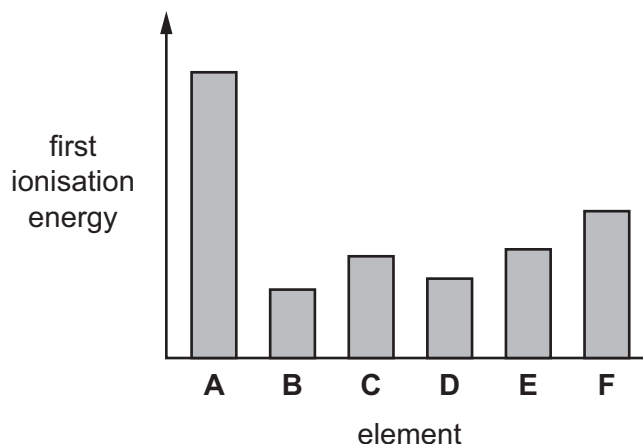
(ii) Explain why sulfur has a lower first ionisation energy than phosphorus.

[2]

54. MAR 18/P22/Q1/a

- 1 (a) The graph shows a sketch of the first ionisation energies of six successive elements in the Periodic Table.

The letters are **not** the symbols of the elements.



- (i) Explain what is meant by the term *first ionisation energy*.

[3]

- (ii) Suggest why the first ionisation energy of **B** is much less than that of **A**.

[3]

55. MAR 18/P22/Q2/d(ii)

- 2 Carbon and silicon are elements in Group 14.

(d) Silicon shows the same kind of bonding and structure as diamond.

- (ii) When silicon reacts with magnesium, Mg_2Si forms. Mg_2Si is thought to contain the Si^{4-} ion.

State the full electronic configuration of the Si^{4-} ion.

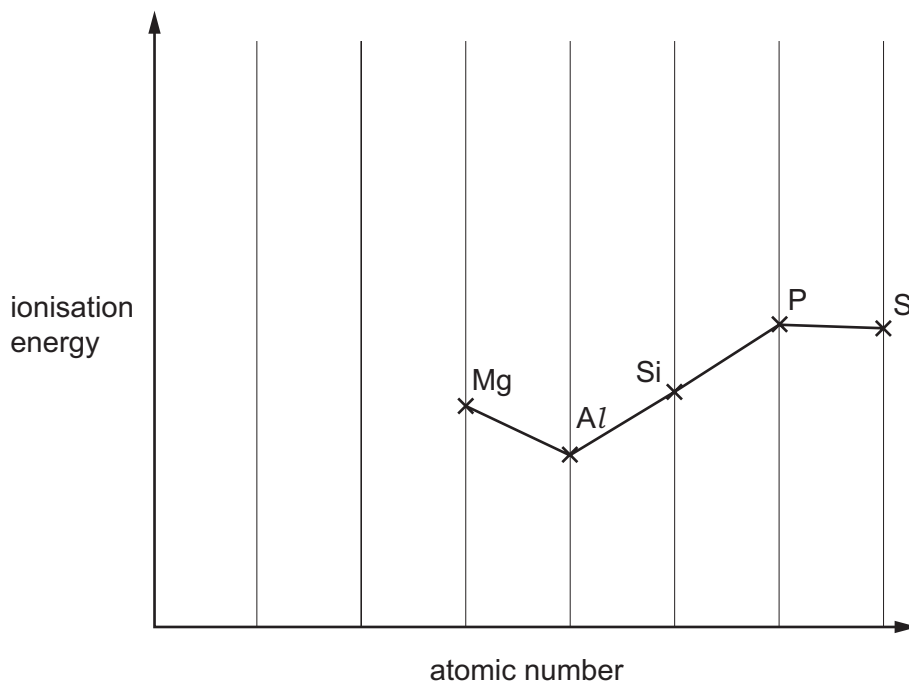
$1s^2$

[1]

56. O/N 17/P21/Q2/a

- 2 The elements in the third period, and their compounds, show trends in their physical and chemical properties.

(a) A sketch graph of the first ionisation energies of five successive elements in the third period is shown.



- (i) Explain why there is a general increase in the first ionisation energy across the third period. [2]
- (ii) Sketch, on the graph, the position of the ionisation energies of the two elements that come before Mg in this sequence. [2]
- (iii) Explain, with reference to electron arrangements, the decreases in first ionisation energy between Mg and Al and between P and S.

Mg and Al

P and S

[4]

57. O/N 17/P22/Q3/a

- 3 The elements in Group 2 show trends in their properties that are typical of metals. The elements in Group 17 show trends in their properties that are typical of non-metals.

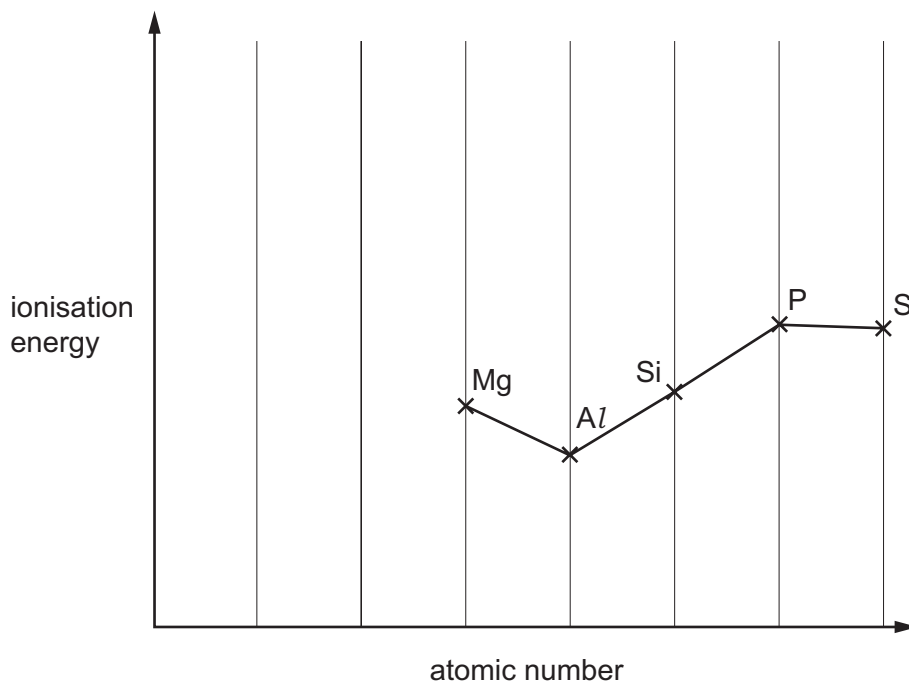
(a) State and explain the trend in ionisation energy down Group 2.

[2]

58. O/N 17/P23/Q2/a

- 2 The elements in the third period, and their compounds, show trends in their physical and chemical properties.

(a) A sketch graph of the first ionisation energies of five successive elements in the third period is shown.



- (i) Explain why there is a general increase in the first ionisation energy across the third period. [2]
- (ii) Sketch, on the graph, the position of the ionisation energies of the two elements that come before Mg in this sequence. [2]
- (iii) Explain, with reference to electron arrangements, the decreases in first ionisation energy between Mg and Al and between P and S.

Mg and Al

P and S

[4]

59. M/J 17/P21/Q3/a(ii,v)

- 3 The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.
- (a) The positions of some elements are indicated. The letters used are **not** the symbols of the elements.

[illegible]

From the elements labelled, give the letter for;

- (ii) the element with the highest first ionisation energy, [1]
- (v) the element that forms the largest cation. [1]

60. M/J 17/P22/Q1/a

- 1 The composition of atoms and ions can be determined from knowledge of atomic number, nucleon number and charge.
- (a) Complete the table.

atomic number	nucleon number	number of electrons	number of protons	number of neutrons	symbol
3		2			${}^6_3\text{Li}^+$
		23	26	32	

[2]

61. MAR 17/P22/Q1/a(iii)

- 1 (a) The table shows information about some of the elements in the third period.

element	Na	Mg	Al	P	S	Cl
atomic radius/nm	0.186	0.160	0.143	0.110	0.104	0.099
radius of most common ion/nm	0.095	0.065	0.050	0.212	0.184	0.181
maximum oxidation number of the element in its compounds	+1					+7

- (iii) The radius of the most common ion of Mg is much smaller than the radius of the most common ion of S.

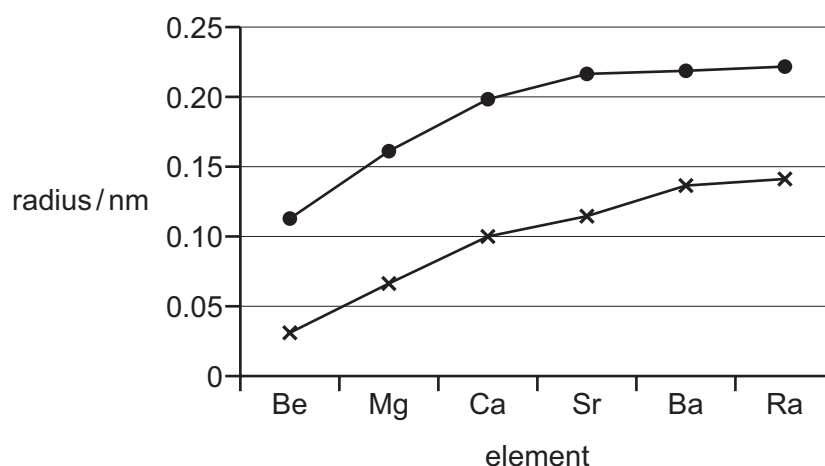
Identify both ions and explain the difference in their radii.

[2]

62. O/N 16/P21/Q3/a

- 3 The elements in Group 2 and their compounds show various trends in their physical and chemical properties.

- (a) The graph below shows the radius values of the atoms and 2+ ions of the elements in Group 2.



- (i) Explain why both lines show a steady increase in the values of the radii down the group.

[2]

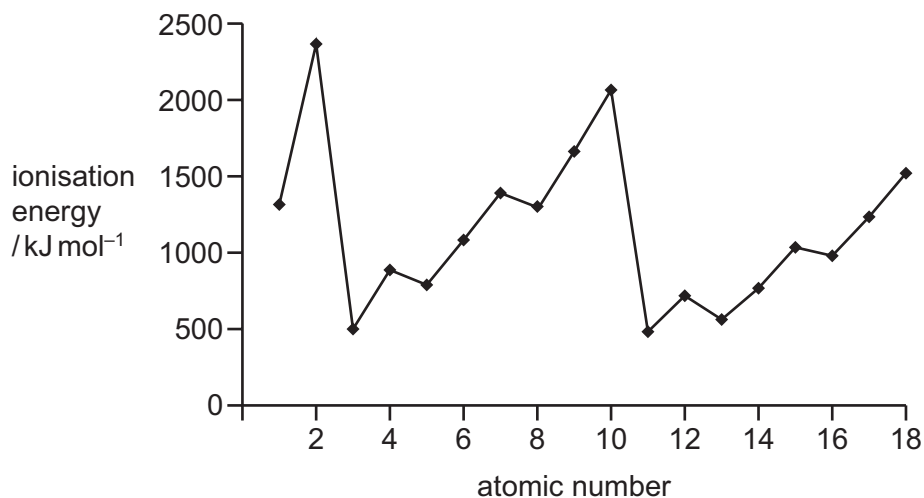
- (ii) State and explain which line represents the atomic radii and which represents the ionic radii.

[2]

63. O/N 16/P22/Q3/a

3 The Periodic Table is arranged such that the properties of the elements show a number of trends.

(a) A plot of the first ionisation energies for the first 18 elements is shown.



(i) Explain why the values show a general increase from atomic number 11 to 18.

[2]

(ii) Explain the decreases in first ionisation energies between

- atomic numbers 12 and 13,
- atomic numbers 15 and 16.

[4]

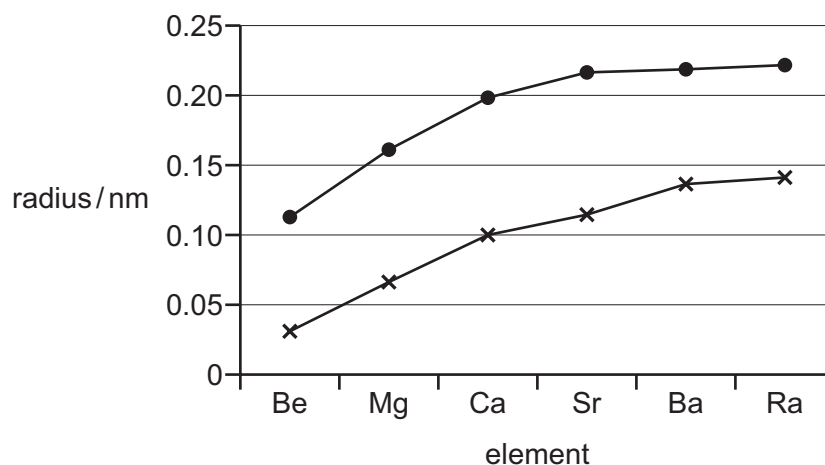
(iii) Suggest an explanation for the trend in the first ionisation energies of the elements with atomic numbers 2, 10 and 18.

[2]

64. O/N 16/P23/Q3/a

- 3 The elements in Group 2 and their compounds show various trends in their physical and chemical properties.

(a) The graph below shows the radius values of the atoms and 2+ ions of the elements in Group 2.



- (i) Explain why both lines show a steady increase in the values of the radii down the group.

[2]

- (ii) State and explain which line represents the atomic radii and which represents the ionic radii.

[2]

65. M/J 16/P21/Q1/a,b,c

- 1 (a) Complete the table to show the composition and identity of some atoms and ions.

name of element	nucleon number	atomic number	number of protons	number of neutrons	number of electrons	overall charge
lithium	6	3				+1
oxygen				9	10	
	54	26	26		24	
			17	18		0

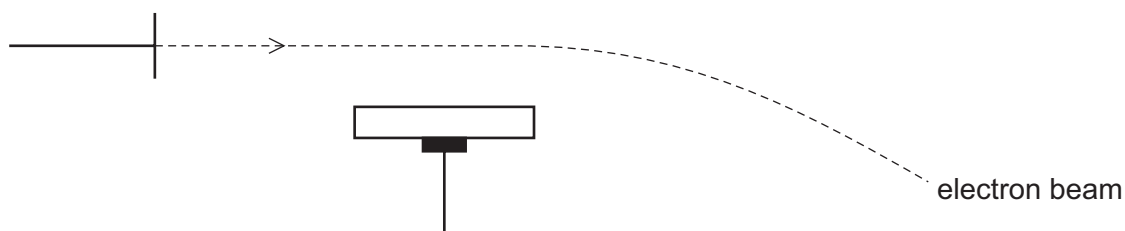
[4]

- (b) Beams of protons, neutrons and electrons behave differently in an electric field due to their differing properties.

The diagram shows the path of a beam of electrons in an electric field.

Add and label lines to represent the paths of beams of protons and neutrons in the same field.





[3]

- (c) The fifth to eighth ionisation energies of three elements in the third period of the Periodic Table are given. The symbols used for reference are **not** the actual symbols of the elements.

	ionisation energies, kJ mol^{-1}			
	fifth	sixth	seventh	eighth
X	6274	21 269	25 398	29 855
Y	7012	8496	27 107	31 671
Z	6542	9362	11 018	33 606

- (i) State and explain the group number of element **Y**.

group number

explanation

[1]

- (ii) State and explain the general trend in **first** ionisation energies across the third period.

[2]

- (iii) Explain why the **first** ionisation energy of element **Y** is less than that of element **X**.

[2]

- (iv) Complete the electronic configuration of element **Z**.

$1s^2$

[1]

66. M/J 16/P22/Q1/a,b

- 1 (a) Complete the table to show the composition and identity of some atoms and ions.

name of element	nucleon number	atomic number	number of protons	number of neutrons	number of electrons	overall charge
boron	10	5				0
nitrogen				8	10	
	208	82	82		80	
			3	3		+1

[4]

- (b) The fifth to eighth ionisation energies of three elements in the third period of the Periodic Table are given. The symbols used for reference are **not** the actual symbols of the elements.

	ionisation energies, kJ mol ⁻¹			
	fifth	sixth	seventh	eighth
X	7012	8496	27 107	31 671
Y	6542	9362	11 018	33 606
Z	7238	8781	11 996	13 842

- (i) State and explain the group number of element **Y**.

group number

explanation

[1]

- (ii) State and explain the general trend in **first** ionisation energies across the third period.

[2]

- (iii) Complete the electronic configuration of element **X**.

1s²

[1]

67. MAR 16/P22/Q1/a(ii)

- 1 This question is about Period 3 elements and their compounds.

- (a) Give an explanation for each of the following statements.

- (ii) The first ionisation energy of sulfur is lower than that of phosphorus.

[2]

68. O/N 15/P22/Q1/a

1 (a) Fill the gaps in the table for each of the given particles.

name of isotope	type of particle	charge	symbol	electron configuration
carbon-13				$1s^2 2s^2 2p^2$
		-1	${}^{37}_{17}\text{Cl}^-$	
sulfur-34	atom	0		
iron-54	cation			$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$

[5]

69. M/J 15/P21/Q1/a

1 (a) Chemists recognise that atoms are made of three types of particle.

Complete the following table with their names and properties.

name of particle	relative mass	relative charge
		0
	1/1836	

[3]

70. M/J 15/P22/Q1/a

1 (a) Chemists recognise that atoms are made of three types of particle.

Complete the following table with their names and properties.

name of particle	relative mass	relative charge
		+1
	1/1836	

[3]

71. M/J 15/P23/Q1/a,b

1 Neon is a noble gas.

(a) Complete the full electronic configuration of neon.

$1s^2$

[1]

(b) (i) Explain what is meant by the term *first ionisation energy*.

[3]

(ii) Explain why the first ionisation energy of neon is greater than that of fluorine.

[2]

Answers Section

1. M/J 26/P21/Q1/b

1(b)(i)	 AND label 1s	1	I n in $1s^n$ I spherical I axes I dot-and-cross diagrams
1(b)(ii)	$\text{Cl(g)} \rightarrow \text{Cl}^+(g) + e^{(-)}$	1	A $\text{Cl(g)} - e^{(-)} \rightarrow \text{Cl}^+(g)$ I $\text{Cl(g)} + e^{(-)} \rightarrow \text{Cl}^+(g) + 2 e^{(-)}$
1(b)(iii)	M1 (small) increase in shielding in Cl is outweighed by M2 increased nuclear charge in Cl M3 Cl has greater nuclear attraction OR M1 similar shielding M2 Cl has greater nuclear charge M3 Cl has greater nuclear attraction	ora ora	3 For all 3 marks to be awarded: 'chlorine' needs only to be identified once in M1 or M2 or M3 chlorine can be identified by name / symbol or trend across a period e.g. number of protons increases across the period M1 great(er) = increase = high(er) = more = large(r) etc for shielding: similar = constant = equal = same shielding = screening I electron removed from same (sub) shell I same number of shells I same electron cloud I references to arrangement of electrons in the orbitals I atomic radius references M2 A number of protons for nuclear charge A greater atomic number for greater nuclear charge A Cl has one more proton than S I 'effective' of effective nuclear charge I nucleon number M3 (nuclear) attraction = pull = draws (into nucleus) For M2 and M3 'nuclear' needs only to be mentioned once e.g. greater nuclear attraction may be implied by reference to 'increased attraction due to increased nuclear charge' owtte If M2 and M3 not awarded: Max 1 for greater nuclear attraction linked to greater nuclear charge if species not identified or incorrectly identified

2. M/J 26/P22/Q1/a

1(a)(i)	4 / four	1	BOD ($1s^2 2s^2$) $2p^4$
1(a)(ii)	2p AND 	1	A $2p^4$ for 2p I p alone I shading but it must have a node I axes I orientation I ignore subscripts $p_x p_y p_z$

3. M/J 26/P22/Q2/a,b

2(a)(i)	curve x = electron AND curve y = proton	1	
---------	--	----------	--

2(a)(ii)	particles in y are deflected less because they have larger mass	1	ecf name given for particles in curve y have the heavier mass. Assume 'they' = y = particles of y unless explicitly refer to 'x' in answer A they are heavier A higher mass / charge ratio A greater weight A they have greater momentum A (particles in) x are lighter I 'electrons have negligible mass' (comparison required) I lower charge density I density I protons are larger I kinetic energy R protons have greater charge density
2(b)(i)	(atom) A and (atom) B OR (atom) A and (atom) C	1	A 6 and 17 OR 6 and 14 A carbon and chlorine OR carbon and silicon
2(b)(ii)	(atom) B and (atom) C	1	A 17 and 14 A chlorine and silicon
2(b)(iii)	M1 atom A has (one) fewer (electron) shell(s) (so decreased shielding effect) M2 (so) more attraction of nucleus for outer/valence electrons / outer or valent shell OR In atom C the effect of more protons / increased nuclear charge is less significant than / outweighed by the (increased shielding) effect of the additional shell ora = ✓✓	2	Subject of question is A, so 'it' = A A reverse argument for atom C I subshells / orbitals I incorrect identities of atoms A, B and C M1 A more shells down group shell = energy level = quantum level If number of shells is given, it must be correct (atom A=2 and atom C=3) M1 A atom A has outer electrons in a lower energy shell M1 I distance from the nucleus M1 I reference to one fewer outer/valence shell M2 A more attraction of protons to the outer/valence electrons M2 BOD pull for attraction

4. M/J 26/P23/Q2/a

2(a)	(down the group) M1 (atomic radius) increases OR (atom) volume increases OR Ne < Ar < Kr < Xe M2 (down the group) the number of shells (of electrons) increases	2	shells = energy levels I number of sub-shells increases I nuclear attraction of outer electron decreases I increased shielding effect outweighs the increase in nuclear charge I outer electron further way
------	---	---	--

5. M/J 26/P23/Q5/b(i)

5(b)(i)	1s 2s 2p 3s 3p	1	A any arrangement of 2 pairs and one single electron in 3 p A double headed arrows.
---------	---	---	---

6. M/J 26/P24/Q1/a

1(a)(i)	atom or ion		number of protons	number of neutrons	2	●✓●✓
	^{27}Al		13	14		
	$^{28}\text{Al}^{3+}$		13	15		

1(a)(ii)	1s 2s 2p 3s 3p	1	A single or double headed arrows A arrows in any direction R lines instead of arrows
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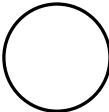
7. MAR 26/P22/Q1/a

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$	1
1(a)(ii)	(ionisation energy decreases because of) - more shells / larger (atomic/ionic) radius - increased shielding - less attraction (for the nucleus/protons) / lower energy required (to remove electrons)	2
1(a)(iii)	$Ba^{2+}(g) \rightarrow Ba^{3+}(g) + e^{-}$	1
1(a)(iv)	M1 electron removed from closer / inner / lower-energy (p sub)shell/orbital M2 (much) less shielding OR greater nuclear attraction	2

8. M/J 25/P21/Q2/a,b(ii),c,d,e

2(a)	atoms (of an element) with same number of protons and different number of neutrons		1
2(b)(ii)			2
	particle	number of particles in one atom of ^{56}Fe	
	protons	26	
	nucleons	56	
2(c)	5		1
2(d)	$\text{Fe(g)} \rightarrow \text{Fe}^{+}(\text{g}) + \text{e}^{(-)}$		1
2(e)	M1 same electronic configuration OR same shielding M2 same number of protons OR same nuclear charge M3 so same nuclear attraction to outer shell electron M4 no change / will not be affected		4

9. M/J 25/P22/Q2/b,c,d,e

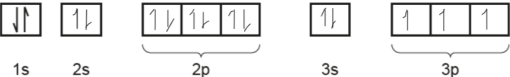
2(b)	1 (pair)	1						
2(c)		1						
2(d)	<table><tr><td>species present in an ion of $^{56}\text{Fe}^{3+}$</td><td>number</td></tr><tr><td>electrons</td><td>M1 23</td></tr><tr><td>M2 neutrons</td><td>30</td></tr></table>	species present in an ion of $^{56}\text{Fe}^{3+}$	number	electrons	M1 23	M2 neutrons	30	2
species present in an ion of $^{56}\text{Fe}^{3+}$	number							
electrons	M1 23							
M2 neutrons	30							
2(e)	M1 smaller / decreases M2 loss of outer / valence shell (of electrons) OR loss of electrons so increase in (force of) attraction from nucleus OR loss of electrons so increase nuclear attraction	2						

10. M/J 25/P23/Q1/a,b

1(a)(i)	$1s^2 2s^2 2p^5$	1
1(a)(ii)	2 (pairs)	1

1(a)(iii)		1
1(b)(i)	$S(g) \rightarrow S^+(g) + e^{(-)}$ OR $S(g) - e^{(-)} \rightarrow S^+(g)$	1
1(b)(ii)	M1 pair of electrons in a 3p orbital (in S) repel each other M2 outweighs the effect of increase in nuclear charge	2
1(b)(iii)	M1 $Na^+ < Ne < F^-$ M2 (all have) same number of electrons / isoelectronic M3 larger species has smaller nuclear charge / fewer protons M4 (so) smaller nuclear attraction (to outer electrons)	4

11. M/J 25/P24/Q1/c

1(c)(i)	$Ar^{2+}(g) \rightarrow Ar^{3+}(g) + e^{(-)}$ M1 a correct equation which represents an ionisation process e.g. $X^{n+} \rightarrow X^{(n+1)+} + e^{(-)}$ M2 correct equation to show third ionisation energy of Ar including (g) state symbols	2
1(c)(ii)	M1 nuclear charge M2 distance of outer electron(s) from nucleus OR ionic radius OR shielding (by inner shells and / or sub-shells)	2
1(c)(iii)	M1 (most significant factor) = distance of outer electron(s) from nucleus OR ionic radius OR shielding OR shells M2 (3rd IE) value for (Na and) Mg is (much) greater than Al (to Ar) OR (Na and) Mg has electron(s) removed from a shell that is closer to the nucleus / less shielding OR (Na and) Mg has electron(s) removed from a shell that is lower in energy	2
1(c)(iv)		1

12. MAR 25/P22/Q2/b,c

2(b)(i)	ionic radius increases / ions larger (down the group) extra shell (of electrons)	2												
2(b)(ii)	ionisation energies increase because less shielding / successive ionic radii are smaller / greater attraction (of electron) to nucleus (large difference between 2nd and 3rd IE) because 3rd e ⁻ coming from an inner shell / lower energy level	2												
2(c)(i)	most of the mass is in the nucleus because protons and neutrons / nucleons are found in the nucleus	2												
2(c)(ii)	<table border="1"> <thead> <tr> <th>isotope</th><th>number of protons</th><th>number of neutrons</th></tr> </thead> <tbody> <tr> <td>magnesium-24</td><td>12</td><td>12</td></tr> <tr> <td>magnesium-25</td><td>12</td><td>13</td></tr> <tr> <td>magnesium-26</td><td>12</td><td>14</td></tr> </tbody> </table>	isotope	number of protons	number of neutrons	magnesium-24	12	12	magnesium-25	12	13	magnesium-26	12	14	2
isotope	number of protons	number of neutrons												
magnesium-24	12	12												
magnesium-25	12	13												
magnesium-26	12	14												
2(c)(iii)	neutron beam horizontal straight line from same source electron beam downwards curving line from same source line deflected more than proton beam	3												

13. O/N 24/P21/Q1/a,b

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
1(a)(ii)		1

1(a)(iii)	3 (three)				1
1(b)	isotope	number of protons	number of neutrons	total number of electron shells	3
	$^{103}_{45}\text{Rh}$	• 45	58	• 5	
	$^{191}_{77}\text{Ir}$	• 77	• 114	6	
	$^{193}_{77}\text{Ir}$	• 77	• 116	6	

14. O/N 24/P21/Q3/c

3(c)(i)	M1 general increase	1
	M2 increase in IE between IE4 and IE5 is noticeably the largest increase between IE's shown	1
3(c)(ii)	$\text{Si}^+(\text{g}) \rightarrow \text{Si}^{2+}(\text{g}) + \text{e}^-$	1

15. O/N 24/P22/Q1/a,b(i)

1(a)(i)	lowest energy state owtte examples • (with) lowest energy • (with) no external energy • (with) no added energy • (where) electrons are not promoted to a higher energy level • most energetically stable	1												
1(a)(ii)	[Ar] <table><tr><td>↑</td><td>↑</td><td>↑</td><td></td><td></td><td>↑↓</td></tr></table> <i>The 3 × d electrons may go in any of the d orbitals</i>	↑	↑	↑			↑↓	1						
↑	↑	↑			↑↓									
1(a)(iii)	12	1												
1(b)(i)	<table><tr><td>isotope</td><td>RIM</td><td># protons</td><td># neutrons</td></tr><tr><td>$^{93}_{41}\text{Nb}$</td><td>92.91</td><td>• 41</td><td>• 52</td></tr><tr><td>$^{181}_{73}\text{Ta}$</td><td>180.95</td><td>• 73</td><td>• 108</td></tr></table>	isotope	RIM	# protons	# neutrons	$^{93}_{41}\text{Nb}$	92.91	• 41	• 52	$^{181}_{73}\text{Ta}$	180.95	• 73	• 108	2
isotope	RIM	# protons	# neutrons											
$^{93}_{41}\text{Nb}$	92.91	• 41	• 52											
$^{181}_{73}\text{Ta}$	180.95	• 73	• 108											

16. O/N 24/P22/Q3/c

3(c)(i)	$\text{P}^{2+}(\text{g}) \rightarrow \text{P}^{3+}(\text{g}) + \text{e}^{(-)}$	1
3(c)(ii)	general increase	1
	increase in IE between IE5 and IE6 is noticeably the largest increase between IEs shown	1

17. O/N 24/P23/Q1/a,b

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
1(a)(ii)		1
1(a)(iii)	3 (three)	1

1(b)	isotope	number of protons	number of neutrons	total number of electron shells	3
	$^{103}_{45}\text{Rh}$	• 45	58	• 5	
	$^{191}_{77}\text{Ir}$	• 77	• 114	6	
	$^{193}_{77}\text{Ir}$	• 77	• 116	6	

18. O/N 24/P23/Q3/c

3(c)(i)	M1 general increase	1
	M2 increase in IE between IE4 and IE5 is noticeably the largest increase between IE's shown	1
3(c)(ii)	$\text{Si}^+(\text{g}) \rightarrow \text{Si}^{2+}(\text{g}) + \text{e}^-$	1

19. M/J 24/P22/Q1/a,b

1(a)	<table border="1"><tr><td>Mg²⁺</td><td>24</td><td>12</td><td>10</td></tr><tr><td>Al³⁺</td><td>27</td><td>13</td><td>10</td></tr></table>				Mg ²⁺	24	12	10	Al ³⁺	27	13	10	2
	Mg ²⁺	24	12	10									
	Al ³⁺	27	13	10									
4 correct = 2 marks 2 or 3 correct = 1 mark 0 or 1 correct = 0 marks													
1(b)	M1 it / Al ³⁺ is smaller (compared to Mg ²⁺) M2 greater nuclear attraction for remaining electrons M3 same shielding effect AND greater nuclear charge				3								

20. M/J 24/P22/Q3/c

3(c)	full electronic configuration of Fe^{3+} $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$	1
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21. O/N 23/P21/Q1/a,b

1(a)(i)	total # 3p e^-	3	4	5	1
	total # unpaired e^-	3	2	1	1
1(a)(ii)	$\text{P}(\text{g}) \rightarrow \text{P}^+(\text{g}) + \text{e}^{(-)}$				1
1(a)(iii)	1000 (kJ mol^{-1})				1
	(S) less / least / lowest nuclear attraction (to outer electrons) (than P OR Cl) OR nuclear attraction for Cl is stronger (compared to S (and or P))				1
	S less nuclear charge than Cl OR Cl has a greater nuclear charge than S				1
	(S has a greater nuclear charge (of outer electrons) than P BUT) S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion				1
1(b)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$				1
1(b)(ii)	• decreases (from P^{3-} to Cl^-) • increased attraction (of outer e^-) for nucleus owtte • increased nuclear charge • same shielding • ✓•✓				2

22. O/N 23/P22/Q1/b

1(b)(i)	total e ⁻ in s subshell	6	6	6	1
	total e ⁻ in p subshell	8	9	10	1
1(b)(ii)	$\text{Si(g)} \rightarrow \text{Si}^{\text{+}}(\text{g}) + \text{e}^{-}$				1
1(b)(iii)	1060 (is circled)				1
	P / it has greater attraction of nucleus for outer electrons compared to Si ORA				1
	P / it has greater nuclear charge than Si ORA				1
	S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion				1
1(b)(i)(v)	hydrogen chloride / HCl				1
1(b)(v)	non-linear				1

23. O/N 23/P23/Q1/a,b

1(a)(i)	total # 3p e ⁻	3	4	5	1
	total # unpaired e ⁻	3	2	1	1
1(a)(ii)	$\text{P(g)} \rightarrow \text{P}^{\text{+}}(\text{g}) + \text{e}^{(-)}$				1
1(a)(iii)	1000 (kJ mol^{-1})				1
	(S) less / least / lowest nuclear attraction (to outer electrons) (than P OR Cl) OR nuclear attraction for Cl is stronger (compared to S (and or P))				1
	<u>S less nuclear charge than Cl</u> OR <u>Cl has a greater nuclear charge than S</u>				1
	(S has a greater nuclear charge (of outer electrons) than P BUT) S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion				1
1(b)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$				1
1(b)(ii)	- decreases (from P^{3-} to Cl^{-}) - increased attraction (of outer e⁻) for nucleus owtte - increased nuclear charge - same shielding •√•√				2

24. M/J 23/P21/Q1/a,b,c,d

1(a)	<table><tr><td></td><td>nucleon number</td><td>number of neutrons</td><td>number of electrons</td></tr><tr><td>Tellurium-130</td><td>130</td><td>78</td><td>52</td></tr></table>		nucleon number	number of neutrons	number of electrons	Tellurium-130	130	78	52	3
	nucleon number	number of neutrons	number of electrons							
Tellurium-130	130	78	52							
1(b)	1 s	1								
1(c)	Te(g) → Te*(g) + e ⁽⁻⁾	1								
1(d)(i)	<i>any two factors responsible for increase in IE in first 6 IE from:</i> • (increased) nuclear attraction / attraction for nucleus • (decreased) shielding (by sub-shells) (IE 4 to 5) • spin-pair repulsion (IE 1 to 2)	2								

1(d)(ii)	successive ionisation energies of tellurium M1 general increase M2 jumps from 4 to 5 and (largest) 6 to 7	2
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25. M/J 23/P23/Q1/d(iii)

1(d)(iii)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 (4s^0)$	1
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26. MAR 23/P22/Q1/c

1(c)(i)	$H(g) \rightarrow H^+(g) + e^-$	1
1(c)(ii)	H (cannot undergo second ionisation because it only) has one electron / H^+ has no electron	1
1(c)(iii)	$1s^2 2s^2 2p^6 3s^2 3p^2$	1

27. O/N 22/P21/Q1/a(i),b

1(a)(i)	columns 1 & 3 identical	1												
	<table><tr><td>isotope</td><td>No of p's</td><td>No of n's</td><td>No of e's</td></tr><tr><td>^{69}Ga</td><td>31</td><td>38</td><td>31</td></tr><tr><td>^{71}Ga</td><td>31</td><td>40</td><td>31</td></tr></table>	isotope	No of p's	No of n's	No of e's	^{69}Ga	31	38	31	^{71}Ga	31	40	31	
isotope	No of p's	No of n's	No of e's											
^{69}Ga	31	38	31											
^{71}Ga	31	40	31											
	● √ √	1												
1(b)(i)	they have the same electron arrangement / electronic configuration	1												
1(b)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	1												
1(b)(iii)	M1 big increase in IE between first and second M2 second (and third) electron(s) is removed from inner shell OR second (and third) electron(s) is removed from a shell closer to the nucleus OR second (and third) electron(s) has a stronger nuclear attraction ora	2												

28. O/N 22/P22/Q1/b

1(b)	<table> <tr> <td>NH_4^+</td><td>11 p⁺ AND 10 e⁻</td><td>1</td></tr> <tr> <td>CO_3^{2-}</td><td>30 p⁺ AND 32 e⁻</td><td>1</td></tr> <tr> <td>PO_4^{3-}</td><td>47 p⁺ AND 50 e⁻</td><td>1</td></tr> </table>	NH_4^+	11 p ⁺ AND 10 e ⁻	1	CO_3^{2-}	30 p ⁺ AND 32 e ⁻	1	PO_4^{3-}	47 p ⁺ AND 50 e ⁻	1	
NH_4^+	11 p ⁺ AND 10 e ⁻	1									
CO_3^{2-}	30 p ⁺ AND 32 e ⁻	1									
PO_4^{3-}	47 p ⁺ AND 50 e ⁻	1									

29. O/N 22/P23/Q1/a(i),b

1(a)(i)	columns 1 & 3 identical	1												
	<table><tr><td><i>isotope</i></td><td><i>No of p's</i></td><td><i>No of n's</i></td><td><i>No of e's</i></td></tr><tr><td>^{69}Ga</td><td>31</td><td>38</td><td>31</td></tr><tr><td>^{71}Ga</td><td>31</td><td>40</td><td>31</td></tr></table>		<i>isotope</i>	<i>No of p's</i>	<i>No of n's</i>	<i>No of e's</i>	^{69}Ga	31	38	31	^{71}Ga	31	40	31
	<i>isotope</i>		<i>No of p's</i>	<i>No of n's</i>	<i>No of e's</i>									
^{69}Ga	31	38	31											
^{71}Ga	31	40	31											
• √ √	1													
1(b)(i)	they have the same electron arrangement / electronic configuration	1												
1(b)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	1												
1(b)(iii)	M1 big increase in IE between first and second M2 second (and third) electron(s) is removed from inner shell OR second (and third) electron(s) is removed from a shell closer to the nucleus OR second (and third) electron(s) has a stronger nuclear attraction ora	2												

30. M/J 22/P21/Q1/a,d,e(iii)

1(a)	Identify and draw the shape of highest energy orbital of Ca 4s AND 	1
1(d)(i)	number of protons: 12 number of neutrons: 13	1
1(d)(ii)	$1s^2 2s^2 2p^6 3s^2$	1
1(e)(iii)	M1 (magnesium isotopes have) identical chemical properties AND same electron(ic) arrangement / configuration M2 different physical properties AND different number of neutrons	2

31. M/J 22/P22/Q4/b(iii)

4(b)(iii)	[Ar] 	1
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32. M/J 22/P23/Q1/a,b,c,d,e

1(a)	- energy required - when one electron is removed - from each atom in one mole of - gaseous atoms two or three points for one mark, four points for two marks	2
1(b)	Group VI / 16 AND large increase (in IE) after 6th	1
1(c)	M1 reference to spin pair repulsion in (3)p orbital (in S) OR due to repulsion of two electrons in a (3)p orbital (in S) M2 outweighs increased nuclear charge (in S)	2
1(d)	M1 similarity in electronic structure / shielding of Al^{2+} and Na both remove electron from (3)s ¹ / single electron in (3)s (sub-level / orbital) OR Al^{2+} and Na have same electronic configuration OR shielding (of outer electron) is the same M2 greater nuclear charge / number of protons Al^{2+} has greater nuclear charge OR 13p compared to 11p	2

1(e)		atomic no.	nucleon no.	no.of neutrons	electronic arrangement	3
	copper –65	29	65	65 – 29 = 36	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ¹⁰ 4s ¹	
	M1 29 AND 65 M2 nucleon no – atomic no ALLOW ecf from M1 M3 electronic arrangement					

33. MAR 22/P22/Q1/a,b,c

1(a)	$O(g) \rightarrow O^+(g) + e^-$	1
1(b)(i)	increase across period AND increased nuclear attraction for (valence / outer) electrons [1] increase in (positive) nuclear charge / number of protons (in the nucleus) [1] similar shielding (of outer electrons) [1]	3
1(b)(ii)	spin-pair repulsion (of electrons) in (2)p <u>orbital</u> [1] outweighs increased nuclear charge [1]	2
1(c)	$1s^2 2s^2 2p^6 3s^2 3p^1$ [1] greatest jump between 3rd and 4th ionisations [1] indicates three electrons in outer shell [1]	3

34. O/N 21/P22/Q2/a(ii,iii,v)

2(a)(ii)	All 3 points correct scores two marks Any 2 points scores one mark <i>nuclear charge increases OR increasing proton number</i> e.g. 17 / Cl has a greater nuclear charge <i>describe the similarity in shielding between the two elements</i> e.g. they have almost the same shielding <i>describe the overall effect in terms of greater nuclear attraction for (outer) electrons</i> e.g. (outer) electrons are attracted more (strongly) to the nucleus	2
2(a)(iii)	M1 describes the difference between 1st IE of elements 15 and 16 (P and S) in terms of either: spin-pair repulsion (in element 16 / S) OR electron pair repulsion (in element 16 / S) M2 describes the location of the electron pair in the (3)p orbital which repel each other	2
2(a)(v)	M1 (anions have) same number of electrons (but increasing proton number) M2 increasing proton number / nuclear charge AND increasing attraction of nucleus for (outer) electrons OR (outers) electrons attracted more (strongly) to the nucleus AND because of increasing proton number / nuclear charge	2

35. M/J 21/P22/Q1/f(ii)

1(f)(ii)	<i>electronic configuration of Mg^{2+} ($1s^2$) $2s^2 2p^6$</i>	1
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36. MAR 21/P22/Q2/b,g(iii)

2(b)(i)	$Cl(g) + e^- \rightarrow Cl^-(g)$	1
2(b)(ii)	M1: increasing proton number but similar shielding M2: greater attraction of nucleus (for outer / valence electrons)	2
2(g)(iii)	$(1s^2) 2s^2 2p^6 3s^2 3p^5$	1

37. O/N 20/P21/Q1/a,b,c

1(a)	$Mg(g) \rightarrow Mg^+(g) + e^-$	1
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1(b)	M1: distance between nucleus and outer e⁻ increases OR outer electron removed from higher energy shell	3
	M2: increased shielding	
	M3: decreased nuclear attraction	
1(c)	M1: greater nuclear attraction	2
	M2: (2nd / 2s) electron being removed from smaller (ion)	

38. O/N 20/P21/Q3/a(iii)

3(a)(iii)	Li ⁺ is 1s ² H ⁻ is 1s ²	1
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39. O/N 20/P22/Q1/a,b(i,ii,iii),c

1(a)(i)	positive / + on left AND negative / – on right	1								
1(a)(ii)	straight line vertically upwards from the source	1								
1(b)(i)	<table><tr><td>type of orbital</td><td>s</td><td>p</td><td>d</td></tr><tr><td>number of orbitals</td><td>4</td><td>9</td><td>5</td></tr></table>	type of orbital	s	p	d	number of orbitals	4	9	5	3
type of orbital	s	p	d							
number of orbitals	4	9	5							
1(b)(ii)	4s ☐ 3p ☐☐☐ 3s ☐☐ 2p ☐☐☐ 2s ☐☐ 1s ☐☐	2								
1(b)(iii)	5	1								
1(c)	answer in terms of subatomic particles in the nucleus same (number of) protons AND different (number of) neutrons	1								

40. O/N 20/P23/Q1/a,b,c

1(a)	$\text{Mg(g)} \rightarrow \text{Mg}^+(\text{g}) + \text{e}^{(-)}$	1
1(b)	M1: distance between nucleus and outer e⁻ increases OR outer electron removed from higher energy shell	3
	M2: increased shielding	
	M3: decreased nuclear attraction	
1(c)	M1: greater nuclear attraction	2
	M2: (2nd / 2s) electron being removed from smaller (ion)	

41. O/N 20/P23/Q3/a(iii)

3(a)(iii)	Li^+ is $1s^2$ H^- is $1s^2$	1
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42. M/J 20/P21/Q1/a(i)

1(a)(i)	(different) number of neutrons.	1
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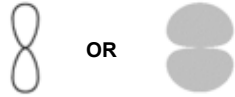
43. M/J 20/P22/Q1/c

1(c)	isotope	nucleon number	total number of electrons in lowest energy level	type of orbital contains the electron in the highest energy level	3
	^{71}Ga	M1 71	M2 2	M3 p (-orbital)	

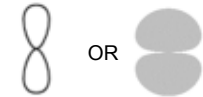
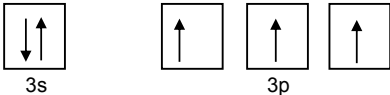
44. O/N 19/P22/Q1/a

1(a)(i)		1
1(a)(ii)	$\text{Si(g)} \rightarrow \text{Si}^+(\text{g}) + \text{e}^-$	1
1(a)(iii)	M1: similar shielding AND increase in proton number / atomic number / nuclear charge M2: increased nuclear attraction	2
1(a)(iv)	M1: 3 OR 13 M2: large(r) <i>increase</i> between third and fourth ionisation energies OR large(r) <i>increase</i> after third electron removed	2

45. M/J 19/P21/Q3/a,b,c

3(a)	$\text{Ar}^+(\text{g}) \rightarrow \text{Ar}^{2+}(\text{g}) + \text{e}^-$ OR $\text{Ar}^+(\text{g}) - \text{e}^- \rightarrow \text{Ar}^{2+}(\text{g})$	1
3(b)	at $x = 8$, within range 13000–20000	1
	at $x = 9$, within range 35000–45000	1
3(c)		1

46. M/J 19/P23/Q1/a,c,d

1(a)(i)	All have the same nucleon number OR same sum / total number of protons + neutrons	1
1(a)(ii)	(different) number of protons, neutrons and electrons	1
1(c)(i)	1s	1
1(c)(ii)		1
1(c)(iii)	M1 3p M2 It is less attracted to the nucleus (so takes less energy to lose) OR It is the highest energy orbital (which is occupied) / it is in the highest energy orbital	2
1(d)(i)		1

1(d)(ii)	M1 (in S, the electron is removed from the) 2 electrons in (3) p orbital OR a pair of electrons in (3) p (orbital / sub-shell) M2 (paired electrons) repel	2
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47. MAR 19/P22/Q2/b(i)

2(b)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
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48. O/N 18/P21/Q1/a(i)

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 (4s^0)$	1
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49. O/N 18/P22/Q1/a,c

1(a)						4
	particle	relative mass	relative charge	location	total number in an atom of ¹⁹⁷ Au	
	electron	0.0005	−1	shell(s) [1]	79	
	neutron	1.(001) [1]	0 [1]	nucleus	118 [1]	

1(c)(i)	M1 (atoms of the same element) with the same proton / atomic number [1] M2 (but) different number of neutrons/mass number [1]	2
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1(c)(ii)	same number of electrons/electronic structure	1
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50. O/N 18/P22/Q2/f

2(f)	M1 process of 'first ionisation energy' involves the loss/removal of an electron [1] M2 Mg and Al AND S and P (in either order) [1] M3 For Al 3p (orbital / sub-level / sub-shell) is higher in energy / further from the nucleus / more shielded (than Mg) [1] <i>ora</i> M4 For S the pair of electrons in the (3) p-orbital repel [1] <i>ora</i>	4
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51. O/N 18/P22/Q4/e(ii,iii)

4(e)(ii)	$1s^2 2s^2 2p^6 (3s^0)$	1
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4(e)(iii)	ions/elements have more shells / energy levels (as the group is descended)	1
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52. O/N 18/P23/Q1/a(i)

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 (4s^0)$	1
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53. M/J 18/P21/Q3/a(ii,iii),b

3(a)(ii)	(ions of Na to Si have) lost outer shell / outer electrons OR atoms have one more shell than (corresponding) ions OR effective nuclear charge is greater for the ion	1
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3(a)(iii)	(P to Cl form ions by) gaining electrons (to the same outer shell / p sub-shell)	1
	Increased repulsion between electrons in same / outer shell / p sub-shell	1

3(b)(i)	(outer) electron removed from 3p subshell / orbital	1
	(3p) higher in energy / more shielded / further from the nucleus	1

3(b)(ii)	(outer) electron for S is paired in a <u>p orbital</u> / S has a full <u>p orbital</u>	1
	causing (spin / electron) pair repulsion (which reduces attraction)	1

54. MAR 18/P22/Q1/a

1(a)(i)	- energy required / energy change - when one electron is removed - from each atom in one mole of - gaseous atoms	max 3
1(a)(ii)	<i>for element B</i> (outer electron is removed) from a higher energy level more shielding less attraction to nucleus	3

55. MAR 18/P22/Q2/d(ii)

2(d)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
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56. O/N 17/P21/Q2/a

2(a)(i)	due to increasing nuclear attraction (for electrons)	1
	due to increasing nuclear charge / atomic / proton number AND similar shielding / same (outer/number of) shell / energy level	1
2(a)(ii)	Cross shown on first vertical line from the y-axis (Group 0 / Ne) is clearly higher than all shown	1
	Cross shown on second vertical line from the y-axis (Group 1 / Na) lower than all shown	1
2(a)(iii)	Al (the outer / valence) electron (which is lost) is in (3)p sub-shell (Mg is in (3)s subshell) OR Al (the outer / valence) electron (which is lost) is in higher energy sub-shell	1 ora
	(electron to be removed) is more shielded / experiences greater screening effect	1 ora
	S has a pair of electrons in (a) (3)p <u>orbital</u> / (a 3)p <u>orbital</u> is full	1 ora
	electron pair repulsion	1

57. O/N 17/P22/Q3/a

3(a)	(IE) <u>decreases</u> / <u>lower</u> because increasing distance of outer electron(s) from nucleus OR increasing distance of outer / valence shell from nucleus OR increased shielding / screening (from inner shells)	1
	reduces nuclear attraction (for electrons)	1

58. O/N 17/P23/Q2/a

2(a)(i)	due to increasing nuclear attraction (for electrons)	1
	due to increasing nuclear charge / atomic / proton number AND similar shielding / same (outer/number of) shell / energy level	1
2(a)(ii)	Cross shown on first vertical line from the y-axis (Group 0 / Ne) is clearly higher than all shown	1
	Cross shown on second vertical line from the y-axis (Group 1 / Na) lower than all shown	1
2(a)(iii)	Al (the outer / valence) electron (which is lost) is in (3)p sub-shell (Mg is in (3)s subshell) OR Al (the outer / valence) electron (which is lost) is in higher energy sub-shell	1 ora
	(electron to be removed) is more shielded / experiences greater screening effect	1 ora
	S has a pair of electrons in (a) (3)p <u>orbital</u> / (a 3)p <u>orbital</u> is full	1 ora
	electron pair repulsion	1

59. M/J 17/P21/Q3/a(ii,v)

3(a)(ii)	H	1
3(a)(v)	F	1

60. M/J 17/P22/Q1/a

1(a)	atomic number	nucleon number	number of electrons	number of protons	number of neutrons	symbol	2 1 1
		6		3	3		
						$^{56}_{26}\text{Fe}^{3+}$	

61. MAR 17/P22/Q1/a(iii)

1(a)(iii)	Mg ²⁺ AND S ²⁻	1
	ion of Mg / Mg ²⁺ has one fewer shell (than ion of S / S ²⁻)	1

62. O/N 16/P21/Q3/a

3(a)(i)	(Atoms/ ions become larger as) the number of (electron) shells increases (down the group)	1	2
	Increased distance of (outer) electrons (from the nucleus) OR Increased shielding results in weaker (nuclear) attraction / pull	1	
3(a)(ii)	top line / dotted line is atomic radii / bottom line / line with crosses is ionic radii (as atoms bigger than ions)	1	2
	Atom has one more shell (than corresponding ion) (ora) OR Atom loses two electrons / outer (shell) electrons / valency electrons (ora) OR Atom loses electrons and so (nuclear) attraction is stronger OR Nuclear charge in ion is greater than the electron(ic) charge (ora) OR Effective nuclear charge in ion is greater (ora)	1	

63. O/N 16/P22/Q3/a

3(a)(i)	Increasing nuclear attraction	1
	Increasing nuclear charge / number of protons AND constant / similar shielding / same shell	1
3(a)(ii)	From 12 / Mg to 13 / Al: (Outer) electron in '13' / Al in (3)p (whereas outer electron in '12' / Mg in (3)s) (3p =) higher energy level / more shielded	1 1
	From 15 / P to 16 / S electron repulsion ('16' / S has a) pair of electrons in a (3)p orbital / a (3)p orbital is full ORA	1 1
3(a)(iii)	(decreasing IE down Group 0) due to decreasing nuclear attraction	1
	increasing shielding / increasing number of shells / energy levels / increasing distance of (outer) electrons (from nucleus)	1

64. O/N 16/P23/Q3/a

3(a)(i)	(Atoms/ ions become larger as) the number of (electron) shells increases (down the group)	1	2
	Increased distance of (outer) electrons (from the nucleus) OR Increased shielding results in weaker (nuclear) attraction / pull	1	
3(a)(ii)	top line / dotted line is atomic radii / bottom line / line with crosses is ionic radii (as atoms bigger than ions)	1	2
	Atom has one more shell (than corresponding ion) (ora) OR Atom loses two electrons / outer (shell) electrons / valency electrons (ora) OR Atom loses electrons and so (nuclear) attraction is stronger OR Nuclear charge in ion is greater than the electron(ic) charge (ora) OR Effective nuclear charge in ion is greater (ora)	1	

65. M/J 16/P21/Q1/a,b,c

1 (a)	name of element	nucleon no.	atomic no.	no. of protons	no. of neutrons	no. of electrons	overall charge		
	lithium	6	3	3	3	2	+1	[1]	[4]
	oxygen	17	8	8	9	10	-2	[1]	
	iron	54	26	26	28	24	+2	[1]	
	chlorine	35	17	17	18	17	0	[1]	
(b)	line straight on labelled 'neutrons' line (curving) up labelled 'protons' proton line clearly shows less (overall) deflection than electron curve							[1] [1] [1]	[3]
(c) (i)	Group 16/6/VI AND Big (owtte) increase/big difference/big gap/big jump/jump in increase/jump in difference after 6th IE							[1]	[1]
(ii)	increases (across period) due to increasing attraction (of nucleus for electrons) due to increasing nuclear charge/atomic/proton number AND constant/similar shielding/same (outer/number of) shell/energy level							[1] [1]	[2]
(iii)	electron (pair) repulsion (Y has a) pair of electrons in a (3)p orbital/a (3)p orbital is full ORA							[1] [1]	[2]
(iv)	$(1s^2)2s^22p^63s^23p^5$							[1]	[1]

66. M/J 16/P22/Q1/a,b

1 (a)	name of element	nucleon number	atomic number	number of protons	number of neutrons	number of electrons	overall charge		
	boron	10	5	5	5	5	0	[1]	[4]
	nitrogen	15	7	7	8	10	-3	[1]	
	lead	208	82	82	126	80	+2	[1]	
	lithium	6	3	3	3	2	+1	[1]	
(b) (i)	Group 17/VII/7 AND big (owtte) increase/big difference/big gap/big jump/jump in increase/jump in difference after 7th IE							[1]	[1]
(ii)	increases across period due to increasing attraction (of nucleus for electrons) due to increasing nuclear charge/atomic/proton number AND constant/similar shielding/ same (outer) shell/energy level							[1] [1]	[2]
(iii)	$1s^22s^22p^63s^23p^4$							[1]	[1]

67. MAR 16/P22/Q1/a(ii)

(ii)	sulfur's electron removed from full (3p) orbital OR sulfur has two electrons in the same orbital electron-electron repulsion (reduces energy required)							[1] [1]	[2]
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68. O/N 15/P22/Q1/a

1 (a)	name of isotope	type of particle	charge	symbol	electron configuration	[5]	[5]
	carbon-13	atom	0	$^{13}_6\text{C}$	$1s^2 2s^2 2p^2$		
	chloride(-37)	anion	1-	Cl^-	$1s^2 2s^2 2p^6 3s^2 3p^6$		
	sulfur-34	atom	0	$^{34}_{16}\text{S}$	$1s^2 2s^2 2p^6 3s^2 3p^4$		
	iron-54	cation	2+	$^{54}_{26}\text{Fe}^{(2+)}$	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$		

69. M/J 15/P21/Q1/a

1 (a)	sub-atomic particle	relative mass	relative charge	[1] [1] [1]	[3]
	neutron	1	0		
	electron	1/1836	-1		
	proton	1	+1		

70. M/J 15/P22/Q1/a

1 (a)	name of particle	relative mass	relative charge	[1] [1] [1]	[3]
	proton	1	+1		
	electron	1/1836	-1		
	neutron	1	0		

71. M/J 15/P23/Q1/a,b

1 (a)	$(1s^2)2s^2 2p^6$	[1]	[1]
(b) (i)	The amount of energy required/energy change when one electron is removed from each atom in one mol of gaseous atoms	[1] [1] [1]	[3]
(ii)	Greater nuclear charge/number of protons Same shielding/number of shells/energy level	[1] [1]	[2]